

电子电路与系统基础(A1)---电阻电路

第9讲：BJT

李国林

清华大学电子工程系

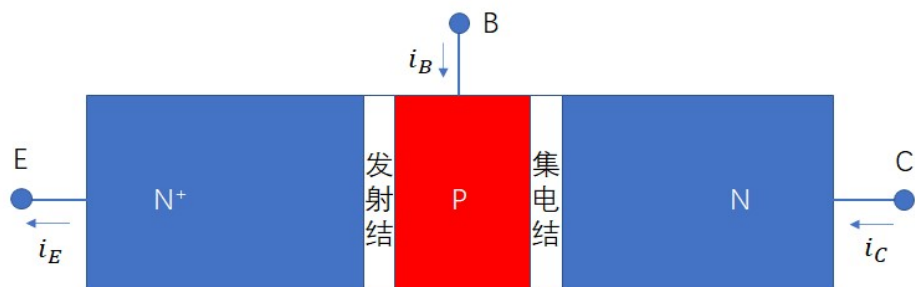
A班课程内容安排

第一学期：电阻	序号	第二学期：动态
电路定律	1	电容电感
电阻电源	2	一阶时域
电路定理	3	正弦稳态
完备方程	4	一阶时频
网络分析	5	二阶时域
二极管	6	二阶时频
晶体管	7	二阶应用
MOSFET	8	寄生效应
BJT	9	频率特性
反相电路	10	作业选讲
数字门	11	负反馈
放大器	12	正反馈
三种组态	13	振荡器
差分放大	14	期末复习
期末复习	15	电路抽象

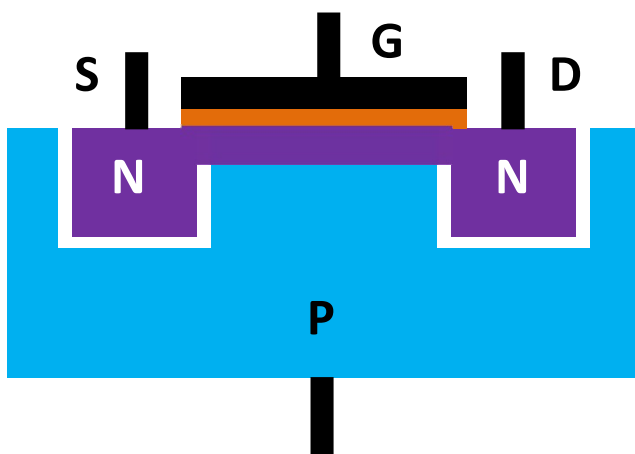
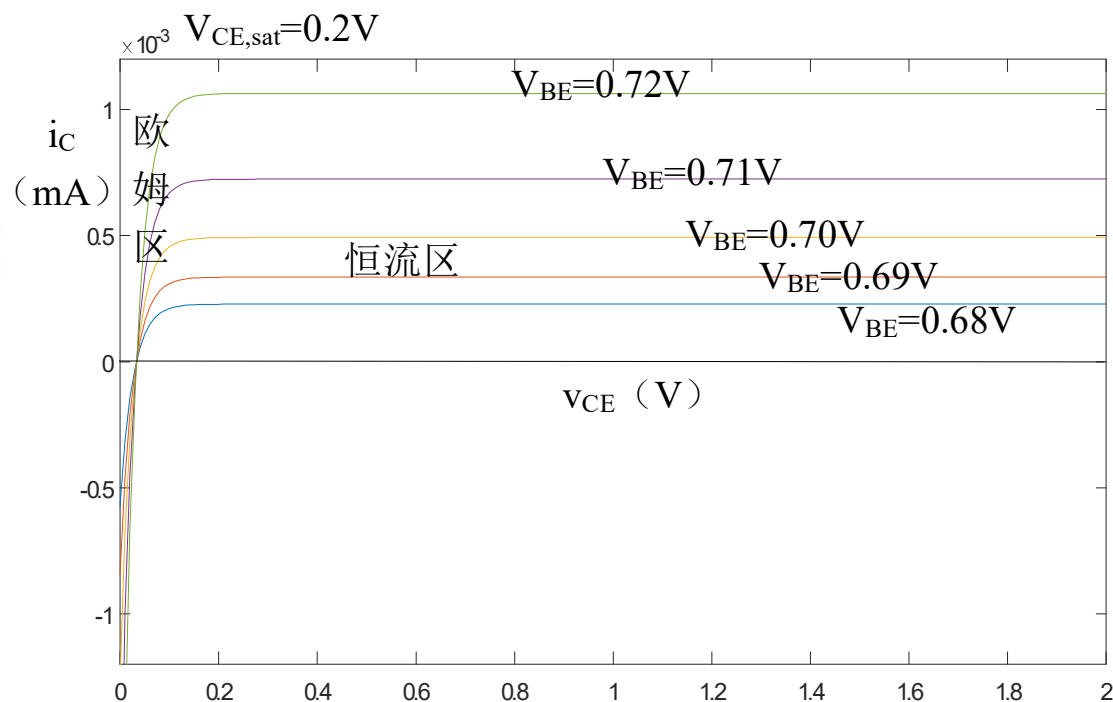
BJT 内容

- 回顾：结构决定特性
- 分段折线电路模型
- 分压偏置电路
 - 负反馈
- 电流镜
- 作业选讲
 - 运放负反馈相关分析

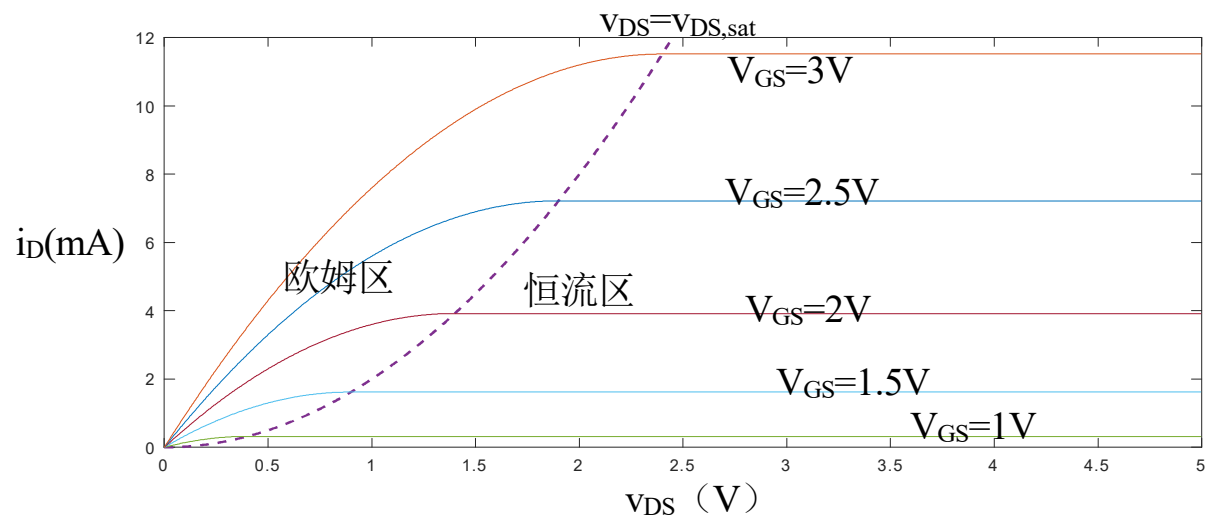
一、控制机制不同，受控电阻特性不同



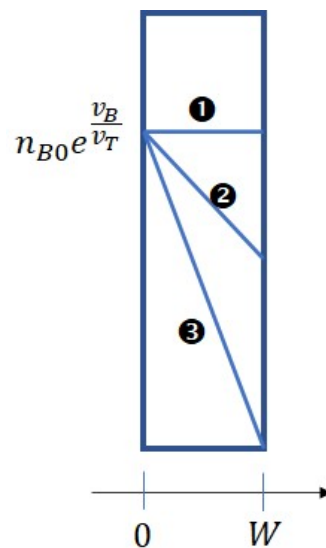
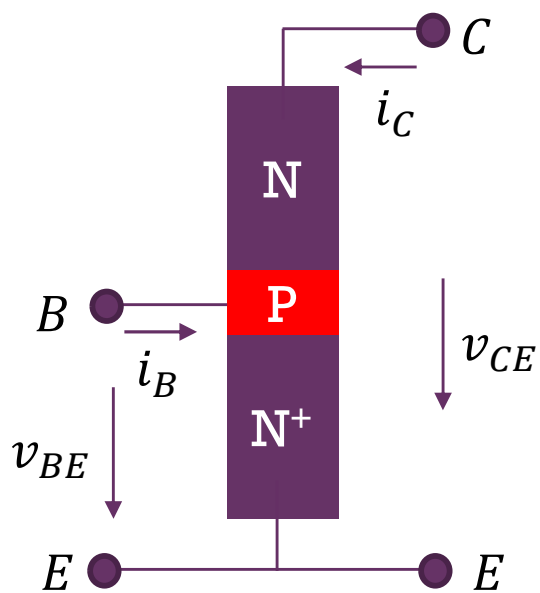
NPN-BJT



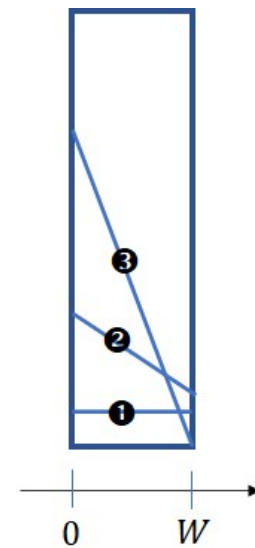
N-MOSFT



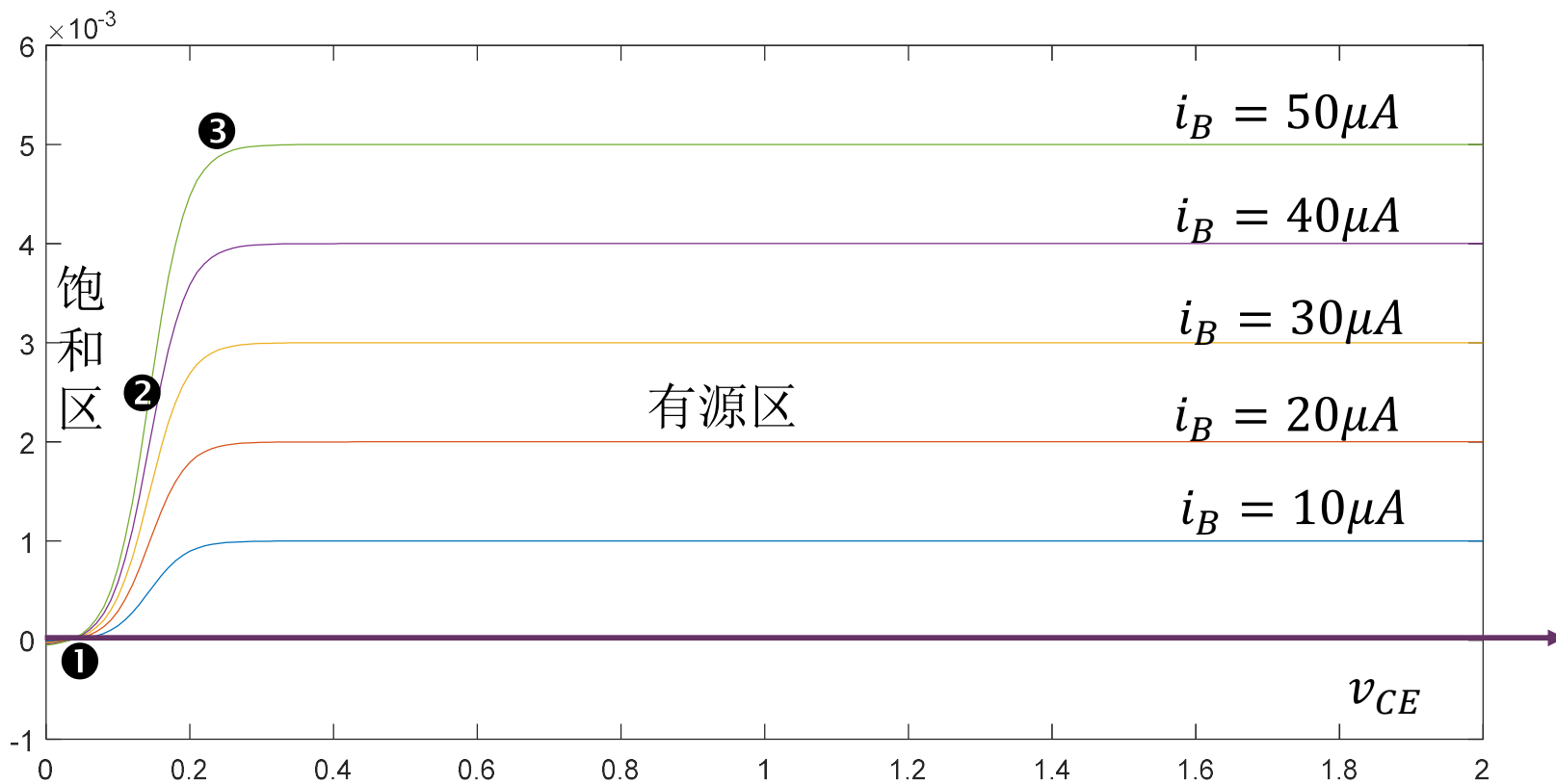
实测多以 i_B 为控制变量



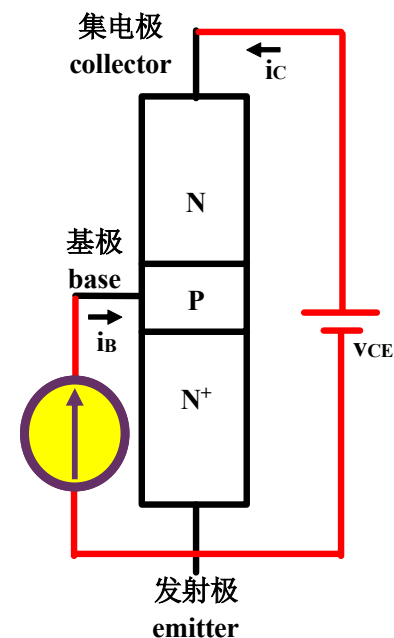
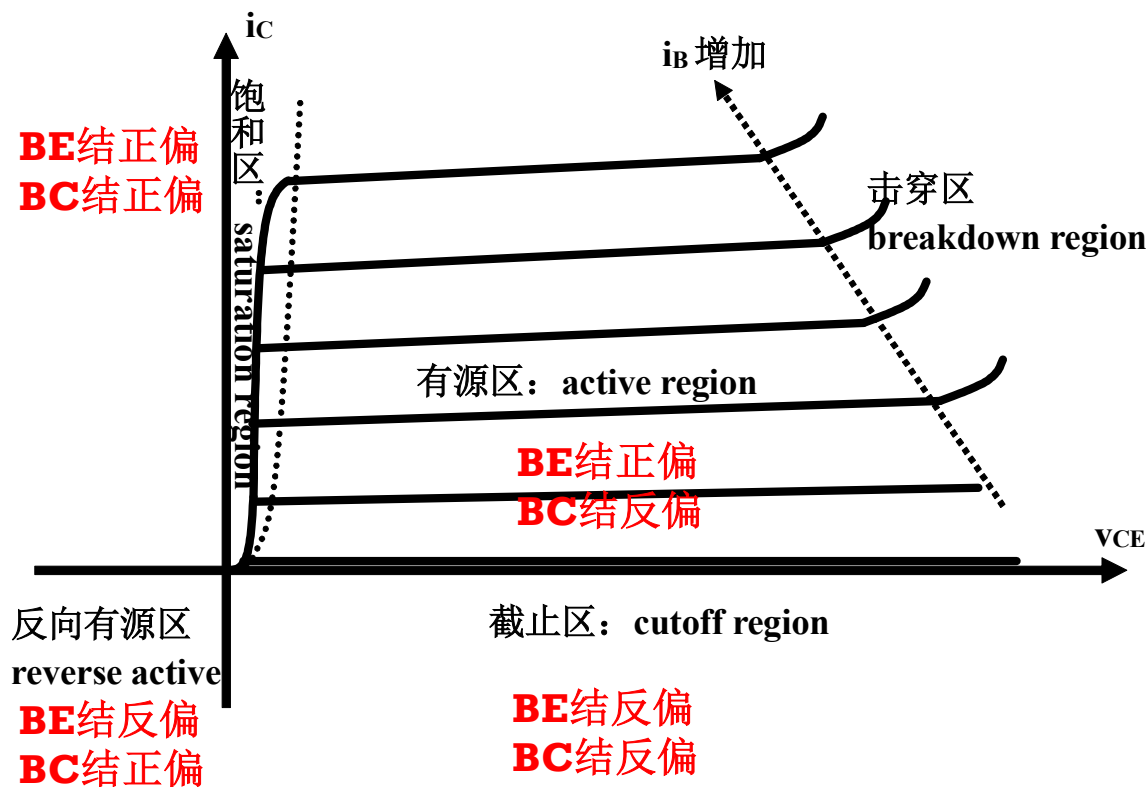
v_{BE} 为控制变量



i_B 为控制变量



牢记恒流区特性方程



$$i_B = A_J J_{BS0} \left(e^{\frac{v_{BE}}{v_T}} - 1 \right)$$

$$i_C = \beta A_J J_{BS0} \left(e^{\frac{v_{BE}}{v_T}} - 1 \right) \left(1 + \frac{v_{CE}}{V_A} \right) = \beta i_B \left(1 + \frac{v_{CE}}{V_A} \right)$$

晶体管位于有源区（恒流区）
的元件约束方程

$$v_T = \frac{kT}{q}$$

热电压

$$V_A$$

厄利电压

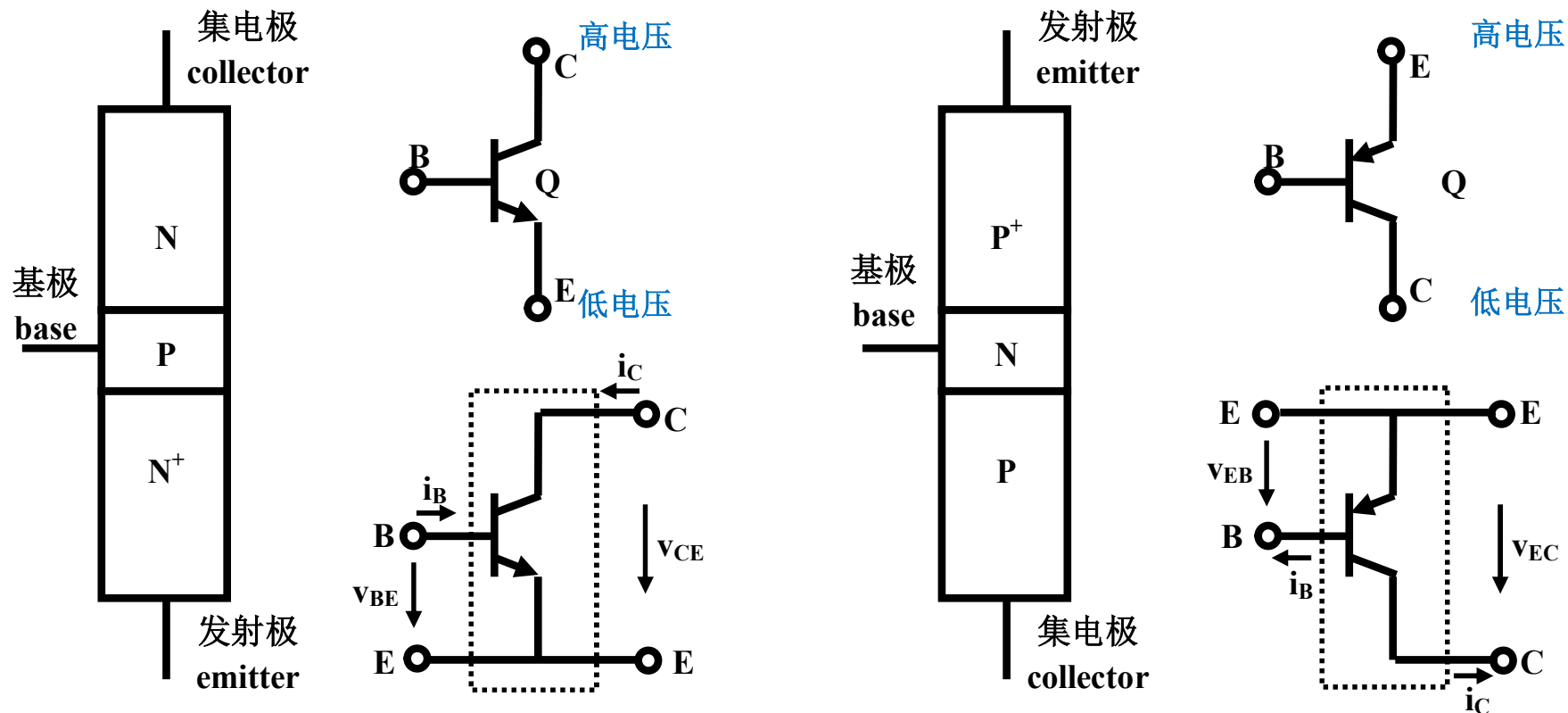
$$I_{BS0} = A_J J_{BS0}$$

反向饱和电流

$$\beta$$

电流增益

PNP和NPN的互补结构



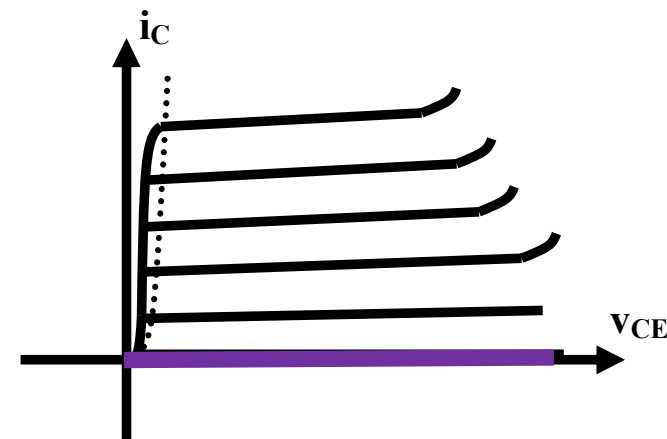
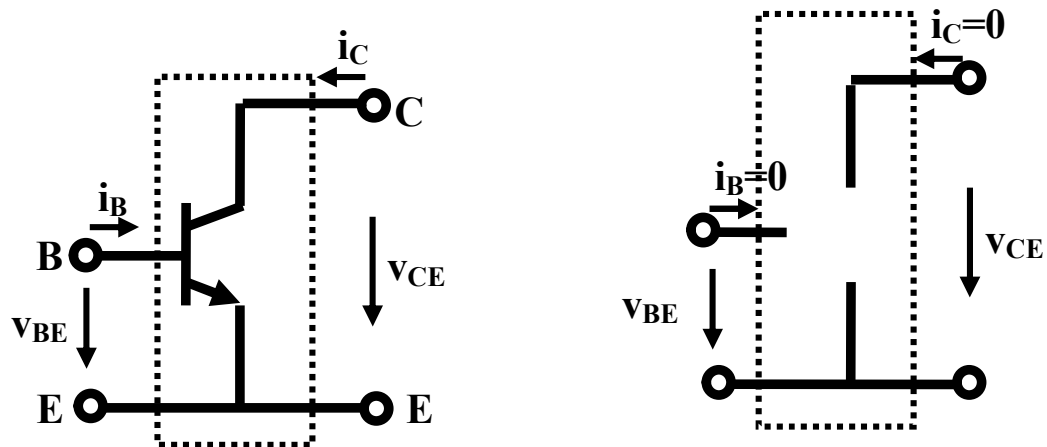
- **PNP-BJT 的元件约束方程和NPN-BJT的形式一致：** PNP方程中，只要将NPN方程中的 V_{BE} 换成 V_{EB} ，将 V_{CE} 换成 V_{EC} ，将 $I_{BS0,n}$ 换成 $I_{BS0,p}$ ，将 β_n 换成 β_p ，将 $V_{A,n}$ 换成 $V_{A,p}$ 后，方程形式没有任何其他变化。

二、分段折线电路模型

- 只要元件约束方程有明显的分区特性，原理性分析即可采用分段折线模型
- BJT伏安特性曲线在第一象限三个分区有明确的物理含义，可在三个区域分别线性化处理
 - 截止区：发射结和集电结均反偏截止：开路模型
 - 饱和区：发射结和集电结均正偏导通，集电极电流剧烈变化下集射电压几乎保持不变，抽象为恒压源
 - 恒流区：发射结正偏，二极管指数律伏安关系；集电结反偏，集电极收集发射区发射到基区的载流子，伏安特性曲线几乎平直，抽象为恒流源

以**NPN-BJT**为例

截止区：开路模型



$$v_{BE} < V_{on} \quad i_B = 0$$

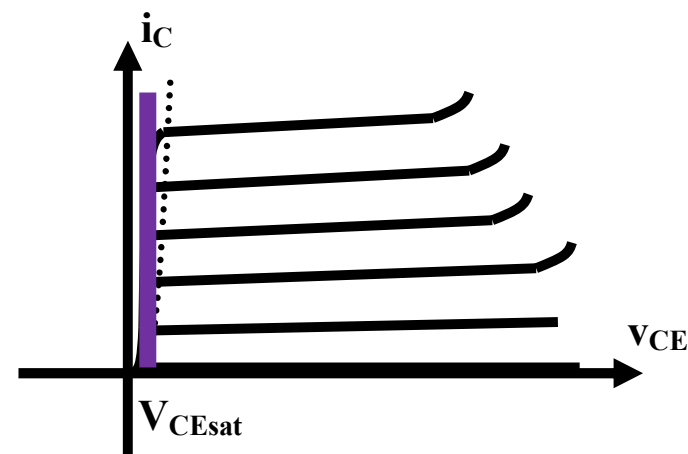
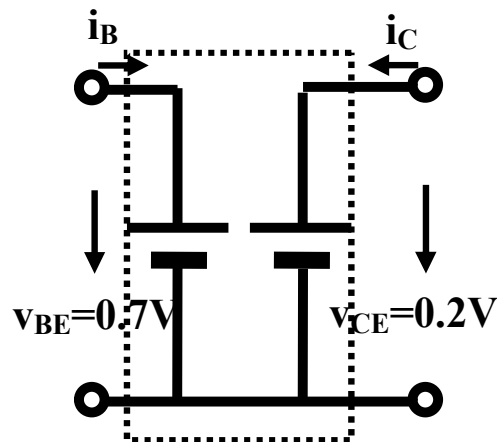
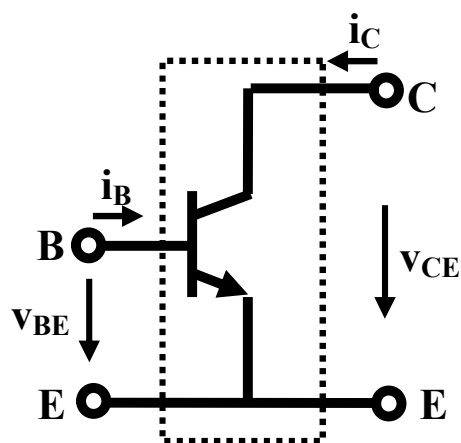
$$i_C = 0$$

$$v_{CE} > 0$$

BE结反偏，截止

BC结反偏，截止

饱和区：恒压模型



$$i_B > 0$$

$$i_C < \beta i_B$$

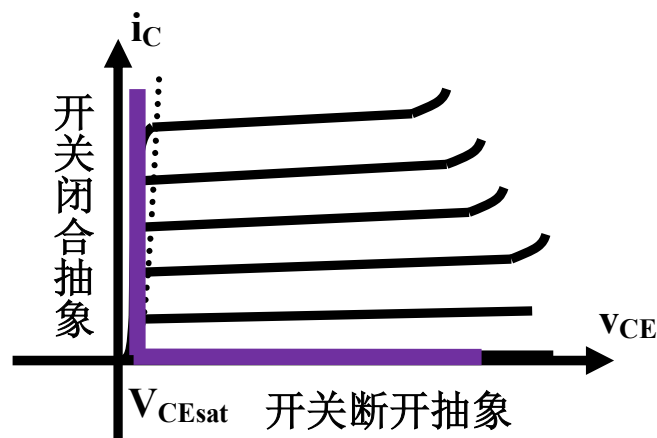
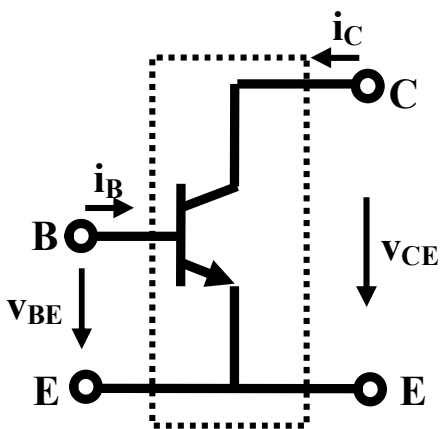
BE结正偏导通

BC结正偏导通

$$v_{BE} = V_{on} = 0.7V$$

$$v_{CE} = V_{CE,sat} = 0.2V$$

截止和饱和：开关模型



$$i_C = 0$$

BE结反偏，**BC**结反偏：开关断开

条件：

$$i_B = 0$$

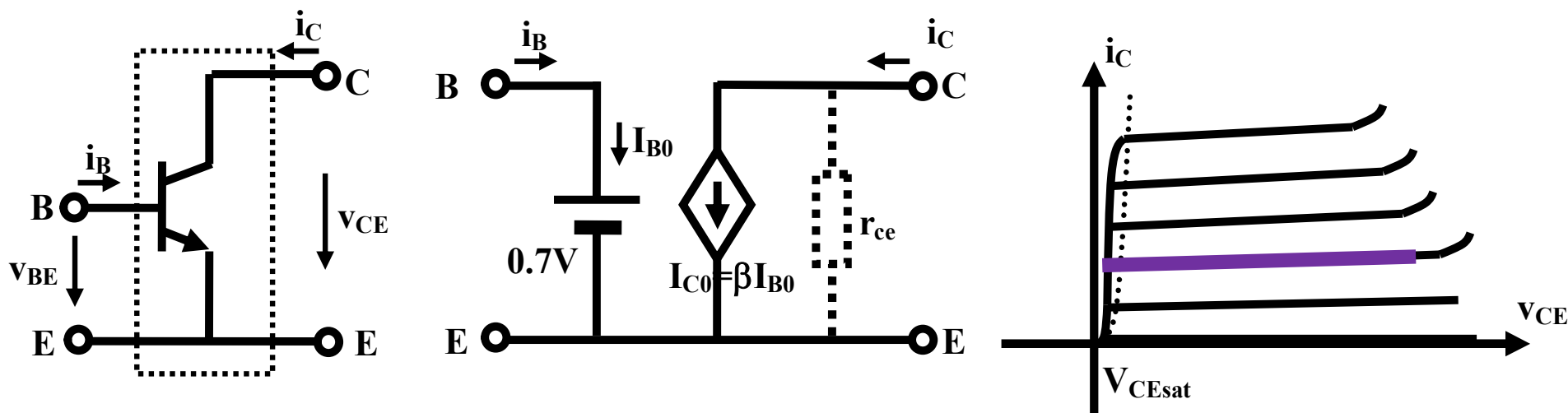
$$v_{CE} = 0.2V$$

BE结正偏，**BC**结正偏：开关闭合

$$i_B \text{ 较大} \quad i_C < \beta i_B$$

很多应用情况下，小的饱和电压被忽略不计，建模为短路

有源区：恒流源模型



$$i_B > 0$$

$$v_{BE} = 0.7V$$

BE结分段折线模型
正偏**0.7V**恒压

BE结正偏，**BC**结反偏

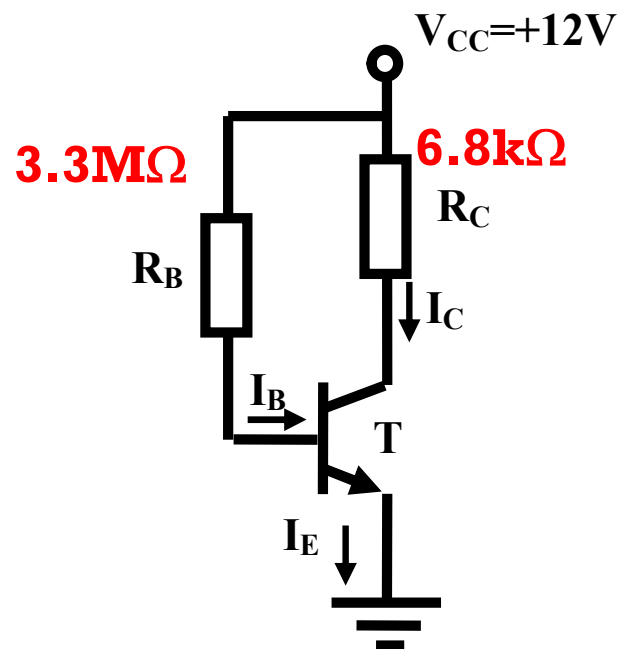
条件： $V_{CE} > V_{CE,sat}$

$$i_C = \beta I_{B0} + \frac{v_{CE}}{r_{ce}} \approx \beta I_{B0}$$

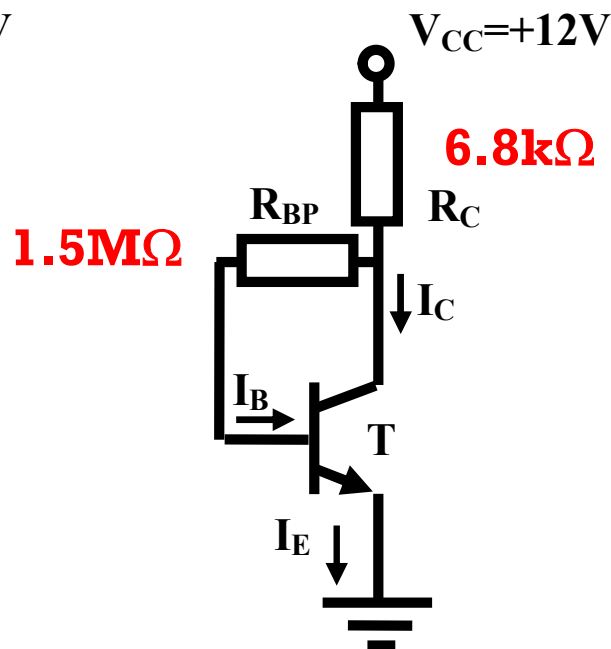
分段折线化模型中，厄利效应大多被忽略
局部线性化分析中，厄利效应一般会加以考虑

$$r_{ce} = \frac{V_A}{I_{C0}}$$

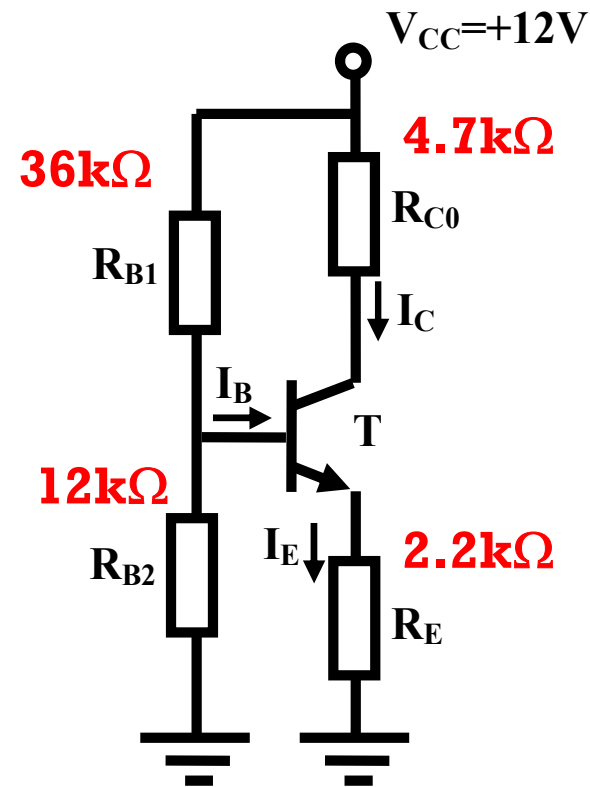
三、分压偏置电路分析



一般偏置



并联负反馈



串联负反馈
分压偏置电路

- BJT工作在恒流区则等效为恒流源
- 恒流区工作的分立BJT，分压偏置电路是最常见的直流偏置电路
 - 由于负反馈的存在，使得直流工作点很稳定
- 分立BJT: $\beta=200\sim450$

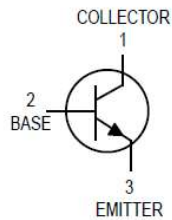
BJT关键参量 β 高度不确定性

MOTOROLA
SEMICONDUCTOR TECHNICAL DATA

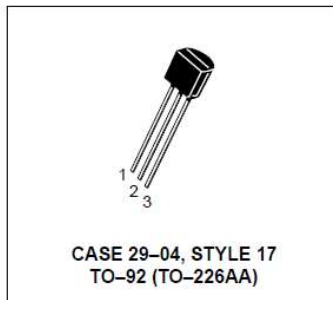
Order this document
by BC549B/D

Low Noise Transistors NPN Silicon

Motorola BC549



DC current gain
versus collector current



BC549B,C
BC550B,C

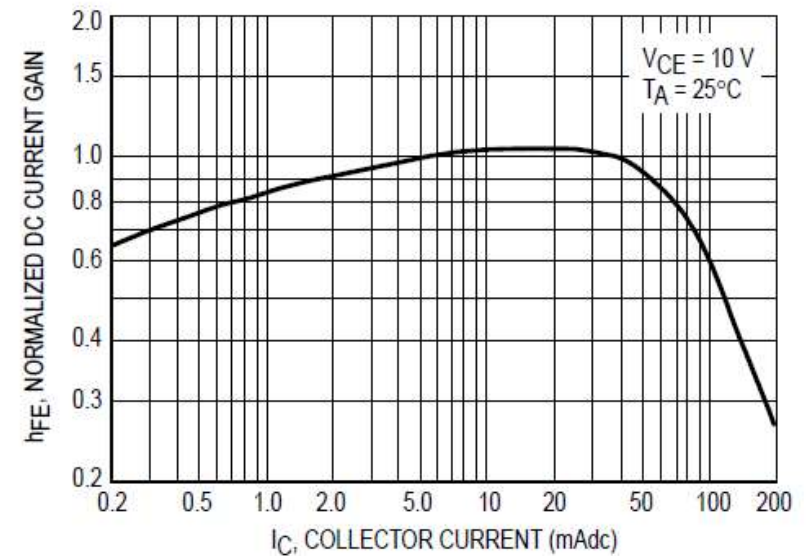
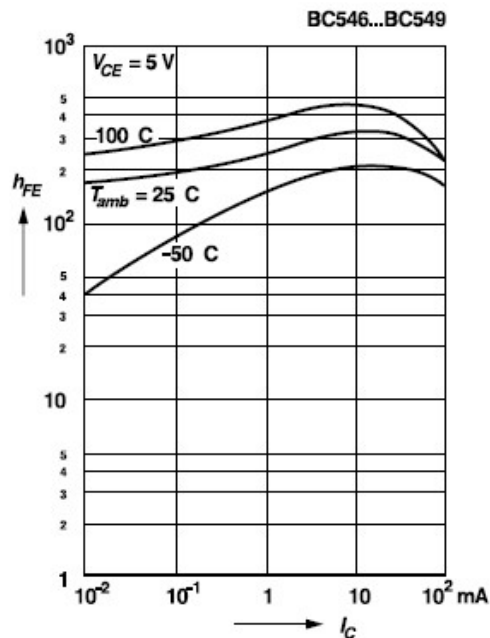


Figure 2. Normalized DC Current Gain

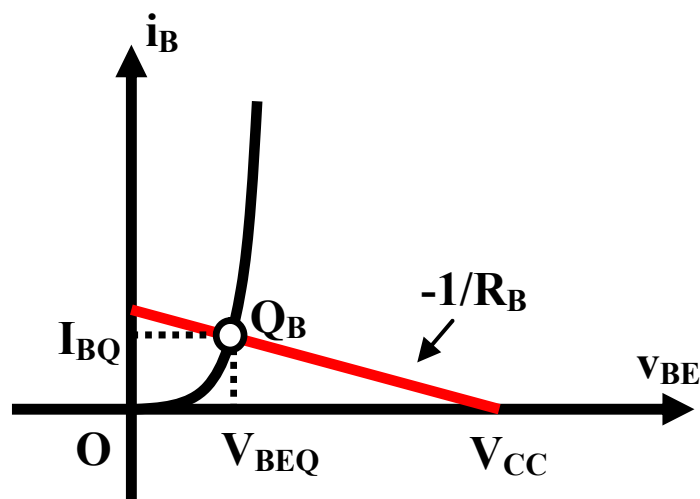
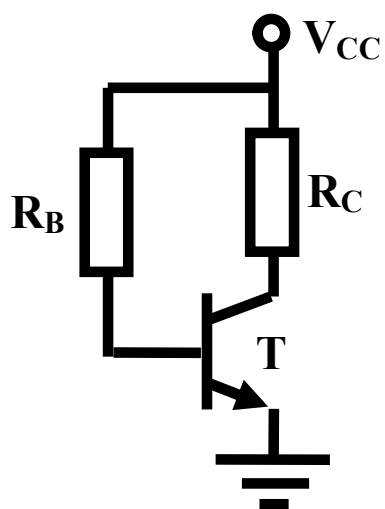
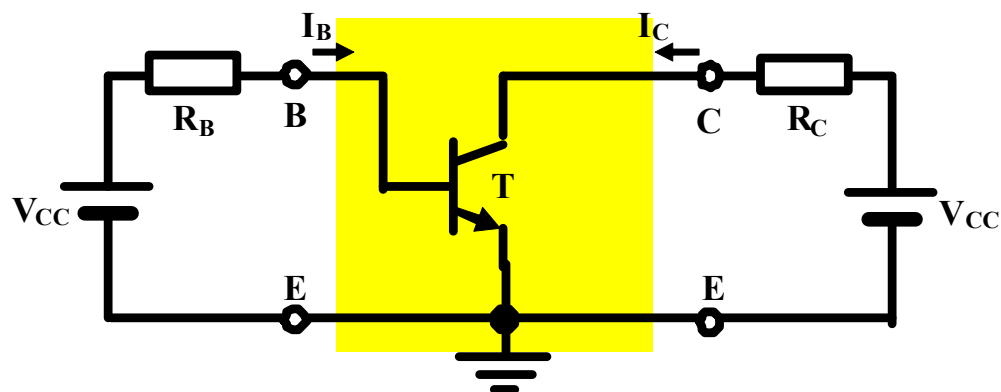
$$\begin{bmatrix} v_{be} \\ i_c \end{bmatrix} = \begin{bmatrix} h_{ie} & h_{re} \\ h_{fe} & h_{oe} \end{bmatrix} \begin{bmatrix} i_b \\ v_{ce} \end{bmatrix} \quad h_{FE} = \frac{I_C}{I_B} \bigg|_{V_{CE}=V_{CE0}}$$

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted) (Continued)

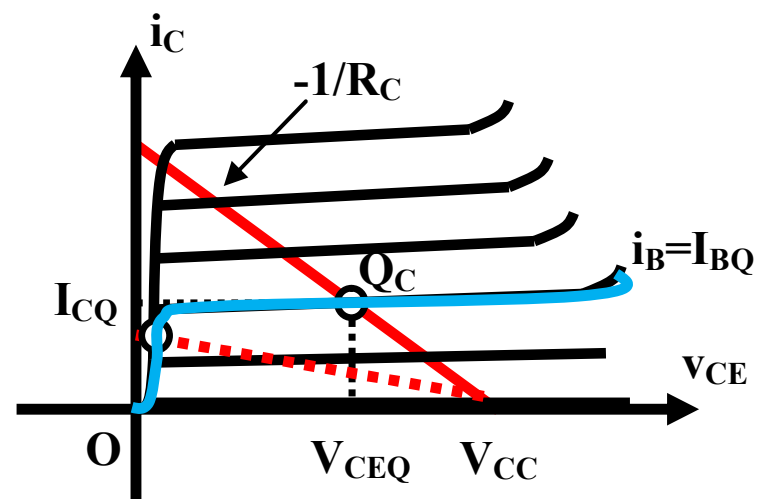
Characteristic	Symbol	Min	Typ	Max	Unit
ON CHARACTERISTICS					
DC Current Gain ($I_C = 10 \mu\text{Adc}$, $V_{CE} = 5.0 \text{ Vdc}$) ($I_C = 2.0 \text{ mAdc}$, $V_{CE} = 5.0 \text{ Vdc}$)	h_{FE}	100 100 200 420	150 270 290 500	— — 450 800	—
Collector-Emitter Saturation Voltage ($I_C = 10 \text{ mAdc}$, $I_B = 0.5 \text{ mAdc}$) ($I_C = 10 \text{ mAdc}$, $I_B = \text{see note 1}$) ($I_C = 100 \text{ mAdc}$, $I_B = 5.0 \text{ mAdc}$, see note 2)	$V_{CE(\text{sat})}$	— — —	0.075 0.3 0.25	0.25 0.6 0.6	Vdc
Base-Emitter Saturation Voltage ($I_C = 100 \text{ mAdc}$, $I_B = 5.0 \text{ mAdc}$)	$V_{BE(\text{sat})}$	—	1.1	—	Vdc
Base-Emitter On Voltage ($I_C = 10 \mu\text{Adc}$, $V_{CE} = 5.0 \text{ Vdc}$) ($I_C = 100 \mu\text{Adc}$, $V_{CE} = 5.0 \text{ Vdc}$) ($I_C = 2.0 \text{ mAdc}$, $V_{CE} = 5.0 \text{ Vdc}$)	$V_{BE(\text{on})}$	— — 0.55	0.52 0.55 0.62	— — 0.7	Vdc



简单偏置图解法分析



BE端口
BE结与戴维南源对接
 输入端口首先确定 I_{B0}



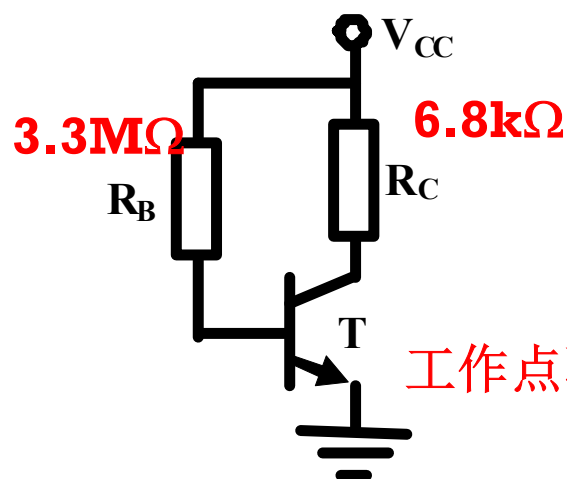
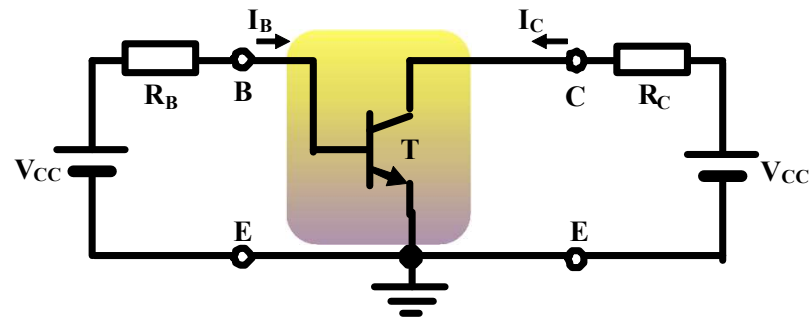
CE端口
 恒流源与戴维南源对接

输出端口再确定 $I_{C0} = \beta \cdot I_{B0}$
 进而确定 V_{CE0}

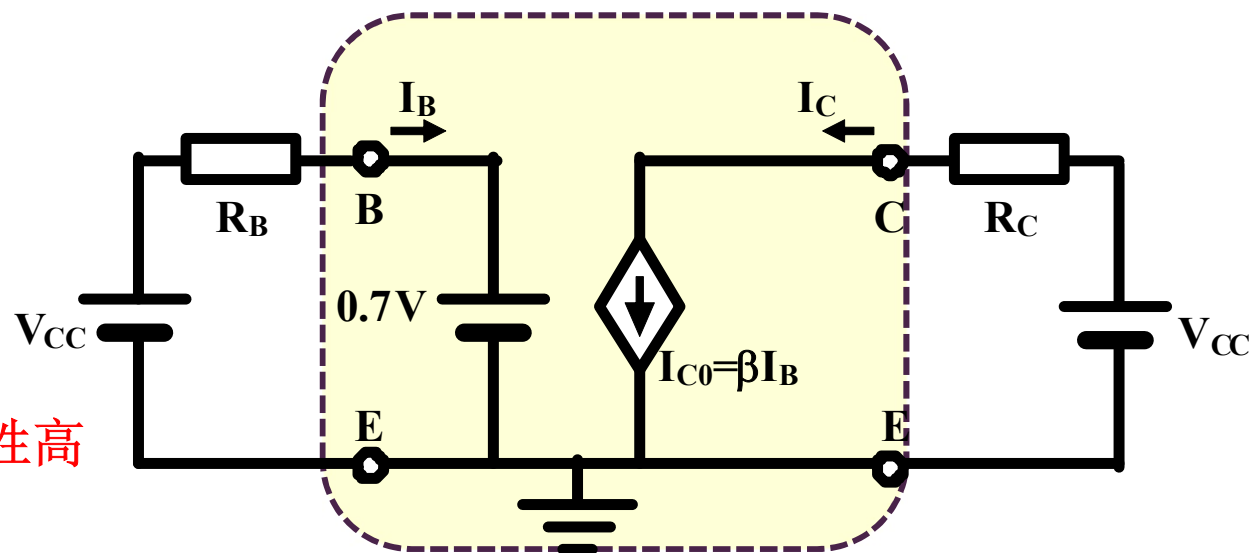
工作区假设

- 当我们不知道晶体管工作在哪个区域时，可以先假设它工作在某个区域，然后按这个区域的电路模型进行计算，如果发现计算结果确实落在这个区域，说明假设正确；如果发现计算结果没有落在假设区域，那么假设就是错误的，需要重新假设工作区域
 - 如果有唯一解，得到合理的解就可以结束
 - 如果有多解，则需每个区域都试试，可能多个假设都成立
- BJT电路分析时，可首先假设其工作在恒流区
 - BJT电路大多属放大器或电流源应用，都要求工作在恒流区
 - 恒流区电流几乎恒定，可等效为电流源：电流源应用
 - 恒流区又称有源区，工作在该区域的晶体管可等效为受控源，具有将直流能量转换为交流能量的能力：放大器、振荡器应用
 - 先假设工作在恒流区，用恒流区电路模型进行分析，检查 V_{CE} 电压，如果大于 V_{CEsat} ，说明确实工作在恒流区，假设成立；如果 V_{CE} 低于 V_{CEsat} ，说明假设不成立，则需重新假设晶体管工作在饱和区，用饱和区电路模型重新分析，分析表明 $I_C < \beta I_{B0}$ ，则可确认晶体管确实工作在饱和区；...

假设工作在恒流区 等效电路



工作点不确定性高

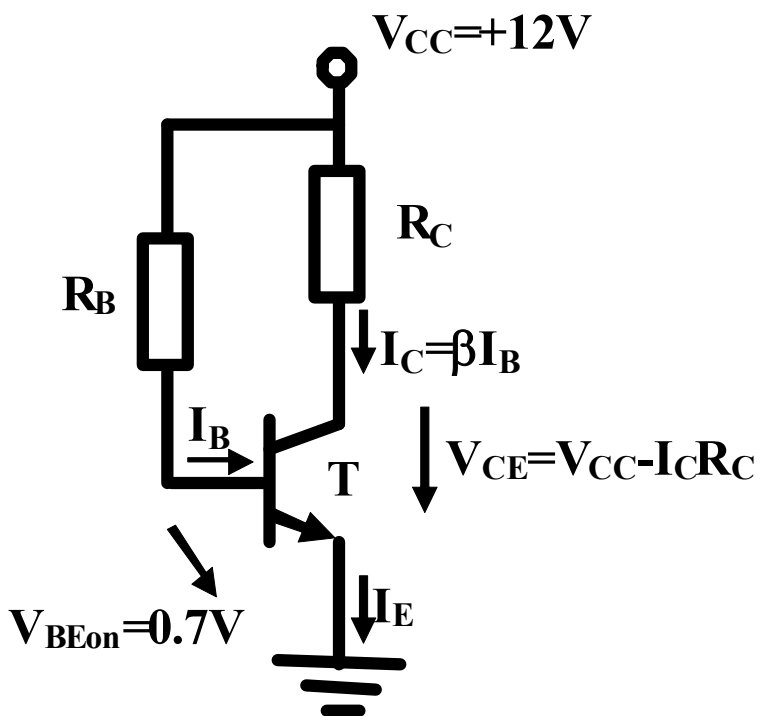


$$I_B = \frac{V_{CC} - V_{BEQ}}{R_B} = \frac{12 - 0.7}{3.3 \times 10^6} = 3.42 \times 10^{-6} = 3.42(\mu A)$$

$$I_C = \beta I_B = (200 \sim 450) \times 3.42 \mu A = 0.685 mA \sim 1.54 mA$$

$$V_{CE} = V_{CC} - I_C R_C = 12 - (0.685 \sim 1.54) \times 6.8 = 7.3V \sim 1.5V > 0.2V = V_{CE,sat}$$

恒流区模型太简单，无需画等效电路



$$V_{BE} = V_{BEon} = 0.7V$$

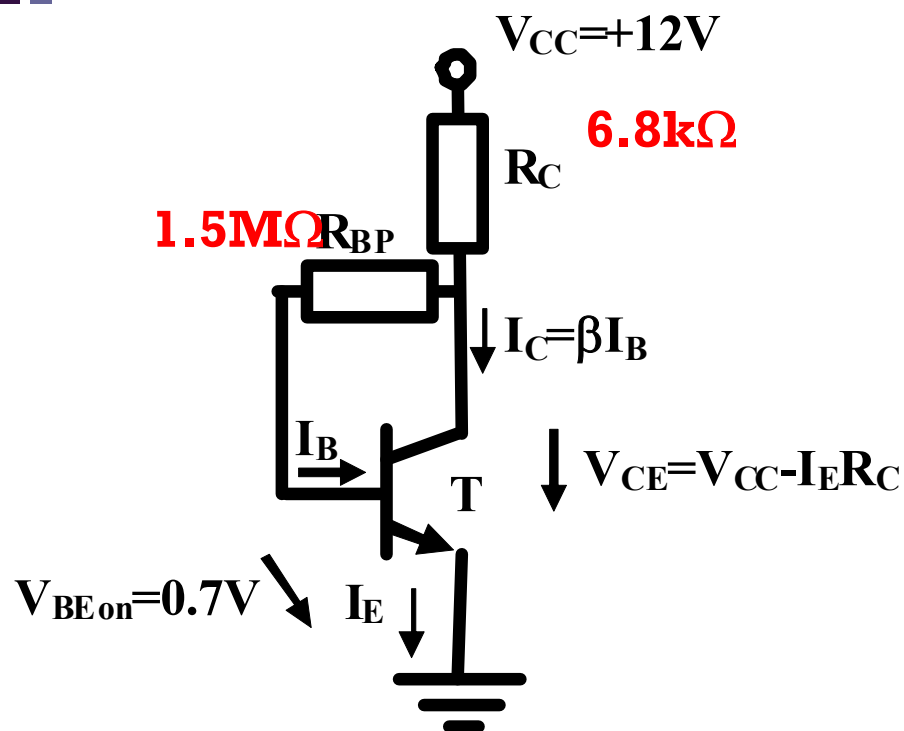
$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B = \beta \frac{V_{CC} - V_{BE}}{R_B}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$= V_{CC} - (V_{CC} - V_{BE}) \frac{\beta R_C}{R_B}$$

并联负反馈



$$V_{CE} = V_{CC} - (\beta + 1)I_B R_C$$

$$= 6.6V \sim 4.4V > V_{CE,sat}$$

可确认在恒流区

并联负反馈使得工作点稳定

$$V_{CC} = (\beta + 1)I_B R_C + I_B R_{BP} + V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_{BP} + (\beta + 1)R_C}$$

$$= \frac{12 - 0.7}{1.5 \times 10^6 + (201 \sim 451) \times 6.8 \times 10^3}$$

$$= 3.94 \mu A \sim 2.47 \mu A \quad \mathbf{1.37 \sim 3.07 M\Omega}$$

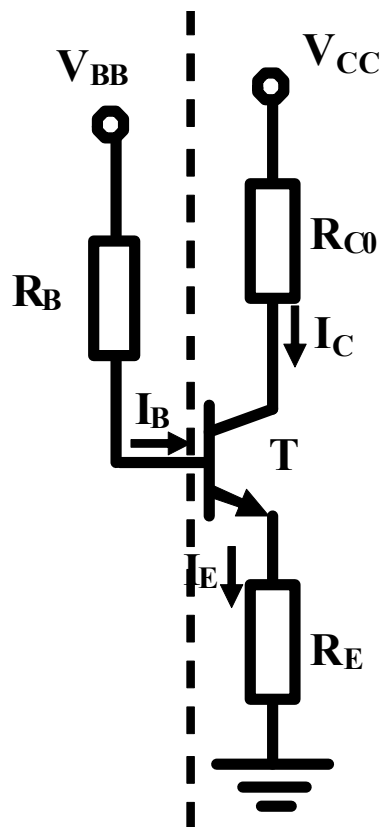
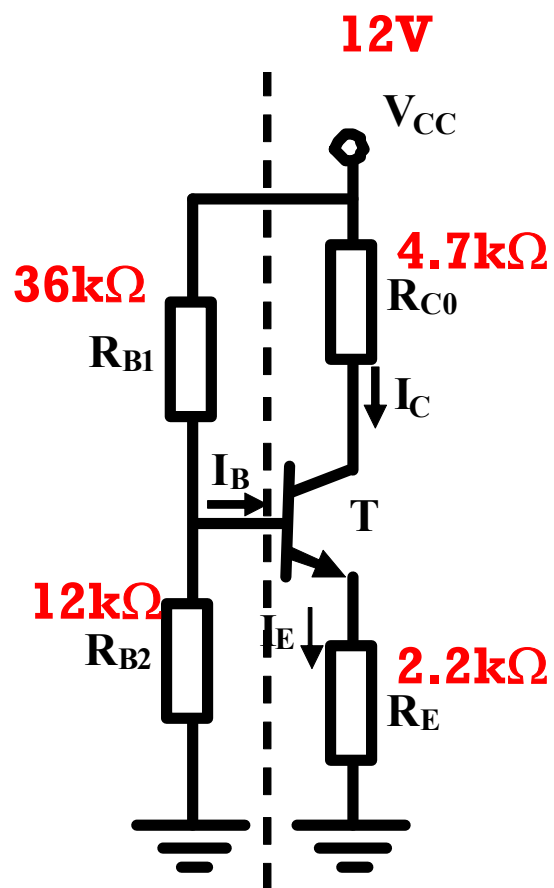
$$I_C = \beta I_B = \beta \frac{V_{CC} - 0.7}{R_{BP} + (\beta + 1)R_C}$$

$$= \frac{V_{CC} - 0.7}{R_C + \frac{R_C + R_{BP}}{\beta}}$$

$$= \frac{12 - 0.7}{6.8 \times 10^3 + \frac{6.8 \times 10^3 + 1.5 \times 10^6}{200 \sim 450}}$$

$$= 0.788 mA \sim 1.11 mA \quad \mathbf{7.5 \sim 3.4 k\Omega}$$

分压偏置电路的戴维南等效



$$V_{BB} = \frac{R_{B2}}{R_{B1} + R_{B2}} V_{CC}$$

$$= \frac{12}{36 + 12} \times 12 = 3V$$

$$R_B = R_{B1} \parallel R_{B2}$$

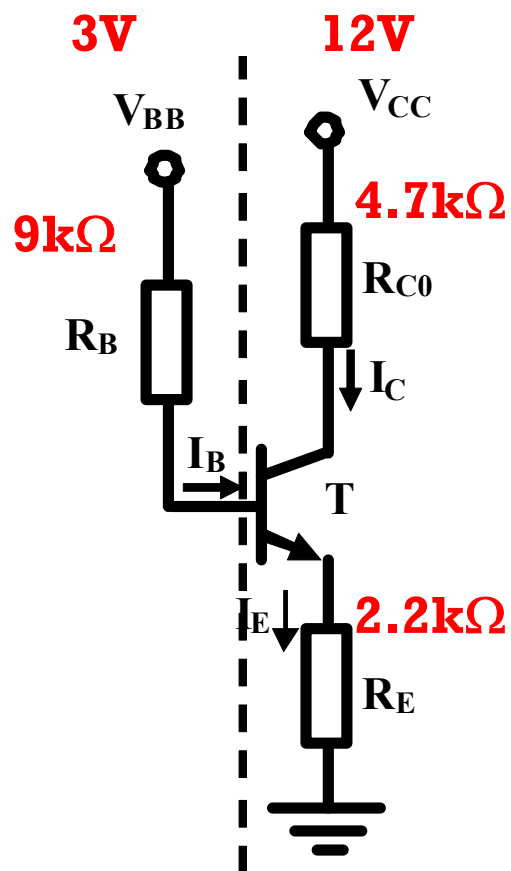
$$= \frac{36 \times 12}{36 + 12} = 9k\Omega$$

串联负反馈

$$V_{BB} = I_B R_B + V_{BE} + (\beta + 1) I_B R_E$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B + (\beta + 1) R_E} = \frac{3 - 0.7}{9 \times 10^3 + (201 \sim 451) \times 2.2 \times 10^3} = 5.10 \mu A \sim 2.30 \mu A$$

442~992kΩ



$$I_C = \beta I_B = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B + R_E}{\beta}} = \frac{3 - 0.7}{2.2 \times 10^3 + \frac{9 \times 10^3 + 2.2 \times 10^3}{200 \sim 450}}$$

$$= \frac{3 - 0.7}{2.2 \times 10^3 + (0.056 \sim 0.025) \times 10^3} = 1.02 mA \sim 1.03 mA$$

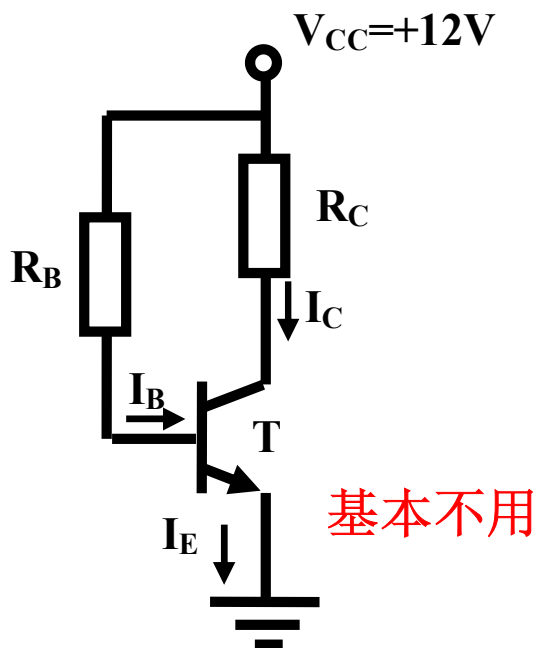
$$V_{CE} = V_{CC} - \beta I_B R_{C0} - (\beta + 1) I_B R_E = 5.0 V \sim 4.9 V$$

可确认在恒流区

串联负反馈使得工作点十分的稳定

工作点确定性/稳定性差异巨大

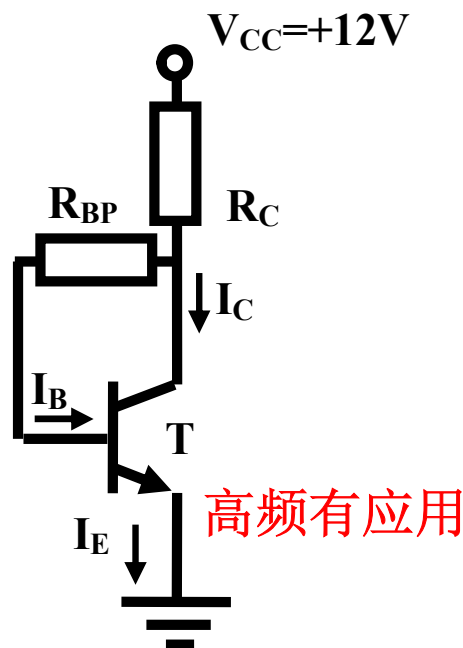
$\beta=200-450$



$$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B}$$

$$= 0.685 \sim 1.54mA$$

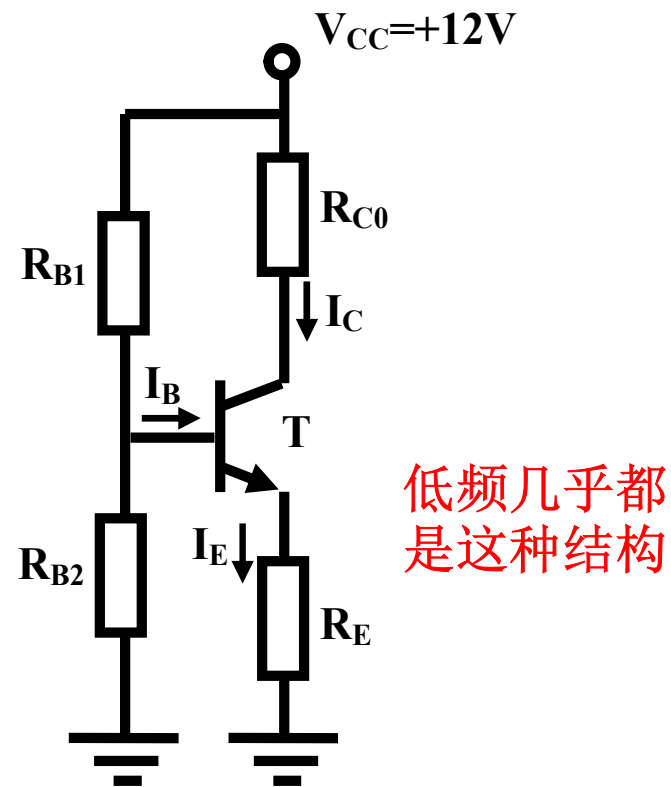
$$V_{CE} = 7.3V \sim 1.5V$$



$$I_C = \frac{V_{CC} - V_{BE}}{R_C + \frac{R_C + R_{BP}}{\beta}}$$

$$= 0.788 \sim 1.11mA$$

$$V_{CE} = 6.6V \sim 4.4V$$

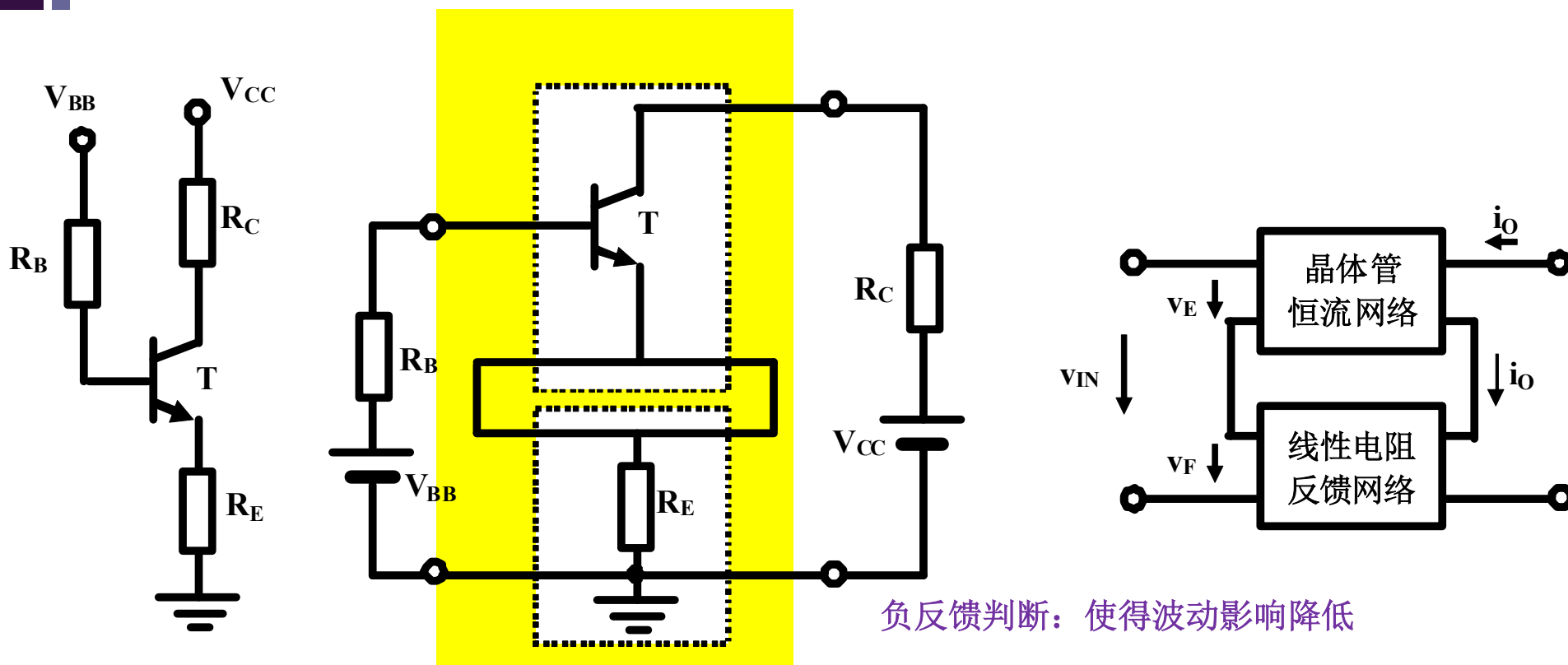


$$I_C = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B + R_E}{\beta}}$$

$$= 1.02 \sim 1.03mA$$

$$V_{CE} = 5.0V \sim 4.9V$$

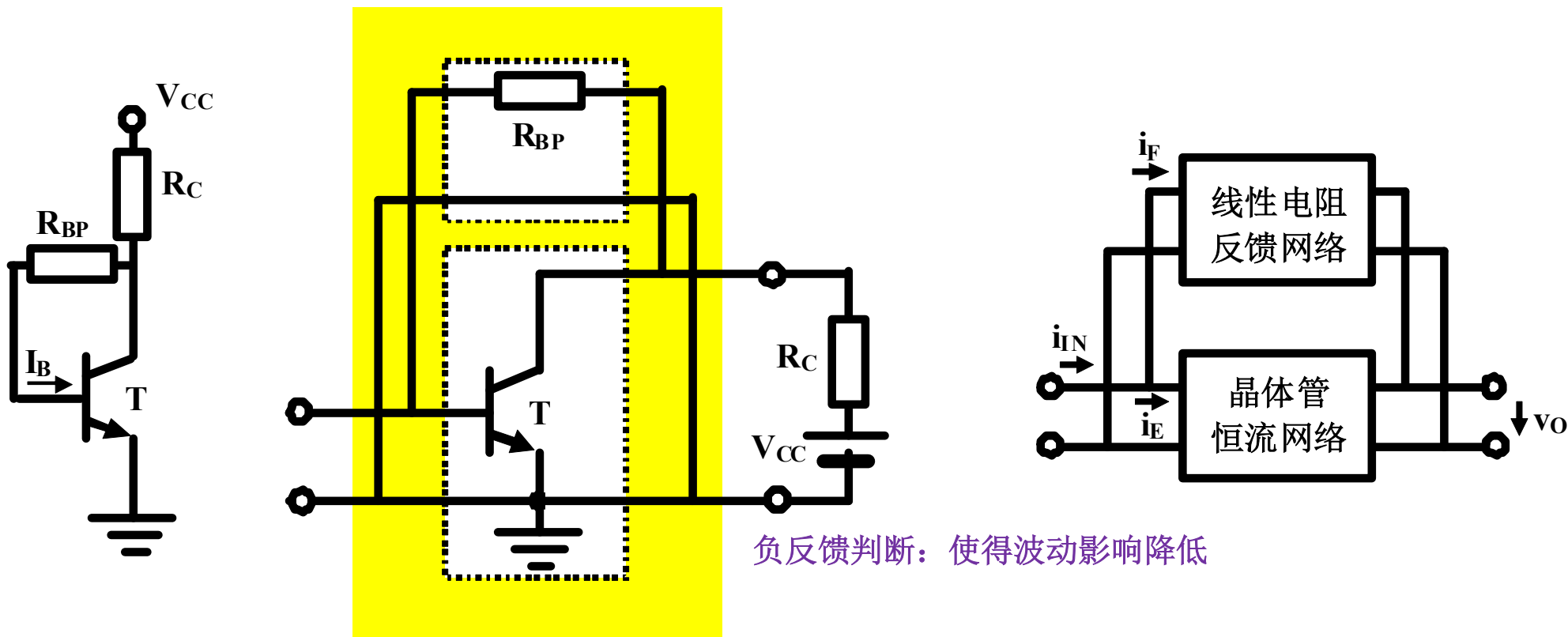
串联负反馈连接：稳定输出电流



串串负反馈，检测输出电流，形成反馈电压
 在负反馈电压作用下，使得输出电流波动降低
 稳定的电流输出，很高的输入和输出阻抗：接近理想压控流源

恒流区晶体管本身就是压控流源模型，串联负反馈适当其时
 使得闭环电路更加接近于理想的压控流源

并联负反馈：稳定输出电压



并并负反馈，检测输出电压，形成负反馈电流
 在负反馈电流作用下，使得输出电压波动降低
 稳定的电压输出，低的输入和输出阻抗：接近理想流控压源

恒流区晶体管本身是压控流源（流控流源）模型，并联负反馈并不十分适当
 并并负反馈形成流控压源受控特性，因而稳定性不如串联负反馈

灵敏度分析

$$S_{x_i}^y = \frac{\Delta y / y}{\Delta x_i / x_i} \stackrel{\Delta x_i \rightarrow 0}{=} \frac{x_i}{y} \frac{\partial y}{\partial x_i}$$

设计值 实际制作偏离设计值

$$y = f(x_1, x_2, \dots, x_n) = f(x_{10} + \Delta x_1, x_{20} + \Delta x_2, \dots, x_{n0} + \Delta x_n)$$

$$= f(x_{10}, x_{20}, \dots, x_{n0}) + \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n + h.o.t$$

$$\Delta y = y - y_0 \approx \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n$$

导致输出偏离设计值

$$\frac{\Delta y}{y_0} \approx \frac{\partial f}{\partial x_1} \frac{x_{10}}{y_0} \frac{\Delta x_1}{x_{10}} + \frac{\partial f}{\partial x_2} \frac{x_{20}}{y_0} \frac{\Delta x_2}{x_{20}} + \dots + \frac{\partial f}{\partial x_n} \frac{x_{n0}}{y_0} \frac{\Delta x_n}{x_{n0}}$$

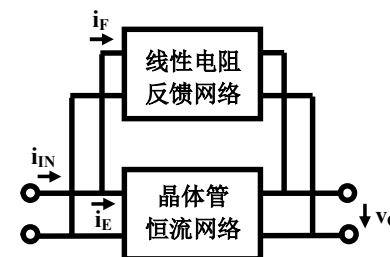
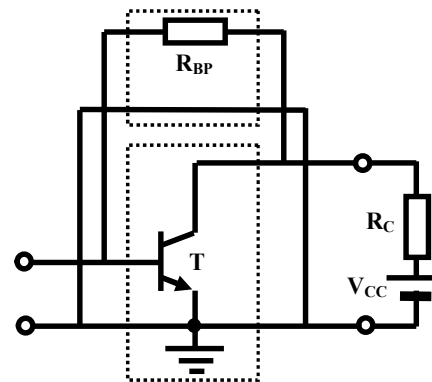
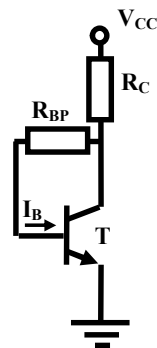
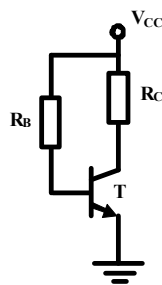
$$\frac{\Delta y}{y_0} = S_{x_1}^y \frac{\Delta x_1}{x_{10}} + S_{x_2}^y \frac{\Delta x_2}{x_{20}} + \dots + S_{x_n}^y \frac{\Delta x_n}{x_{n0}}$$

对于极度不稳定因素如 β ，电路设计时应确保其灵敏度足够小，从而提高系统稳定性

灵敏度：该因素对最终输出的影响力大小

并并负反馈 灵敏度分析

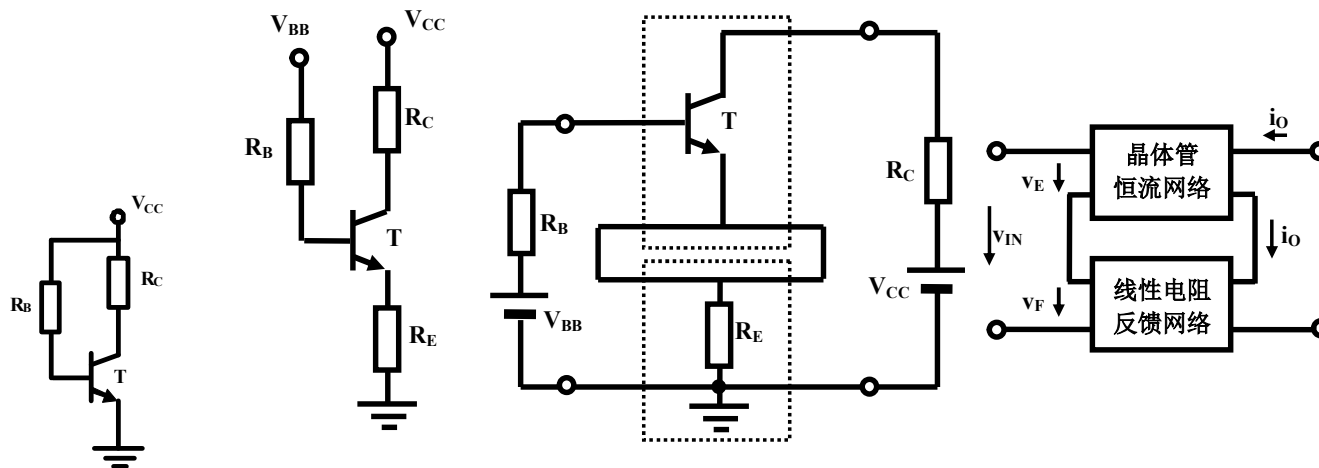
$\beta=300$



	一般偏置	并并负反馈稳定输出电压	并并负反馈
输出电压 v_{ce}	$V_{CE} = V_{CC} - (V_{CC} - V_{BE}) \frac{\beta R_C}{R_B} = 5.01V$	$V_{CE} = \frac{(\beta + 1)R_C V_{BE} + R_{BP} V_{CC}}{(\beta + 1)R_C + R_{BP}} = 5.48V$	
$S_{V_{CC}}^{V_{CE}} = \frac{\partial V_{CE}}{\partial V_{CC}} \frac{V_{CC}}{V_{CE}}$	$= \left(1 - \frac{\beta R_C}{R_B}\right) \frac{V_{CC}}{V_{CE}} = 0.914$	$= \frac{R_{BP}}{(\beta + 1)R_C + R_{BP}} \frac{V_{CC}}{V_{CE}} = 0.926$	
$S_{\beta}^{V_{CE}} = \frac{\partial V_{CE}}{\partial \beta} \frac{\beta}{V_{CE}}$	$= -(V_{CC} - V_{BE}) \frac{R_C}{R_B} \frac{\beta}{V_{CE}} = -1.39$	$= -\frac{V_{CE} - V_{BE}}{\beta + 1 + \frac{R_{BP}}{R_C}} \frac{\beta}{V_{CE}} = -0.502$	
$S_{R_C}^{V_{CE}} = \frac{\partial V_{CE}}{\partial R_C} \frac{R_C}{V_{CE}}$	$= -(V_{CC} - V_{BE}) \frac{\beta}{R_B} \frac{R_C}{V_{CE}} = -1.39$	$= -\frac{V_{CE} - V_{BE}}{R_C + \frac{R_{BP}}{\beta + 1}} \frac{R_C}{V_{CE}} = -0.503$	
$S_{R_B}^{V_{CE}} = \frac{\partial V_{CE}}{\partial R_B} \frac{R_B}{V_{CE}}$	$= (V_{CC} - V_{BE}) \frac{\beta R_C}{R_B^2} \frac{R_B}{V_{CE}} = 1.39$	$= \frac{V_{CC} - V_{CE}}{(\beta + 1)R_C + R_{BP}} \frac{R_{BP}}{V_{CE}} = 0.503$	

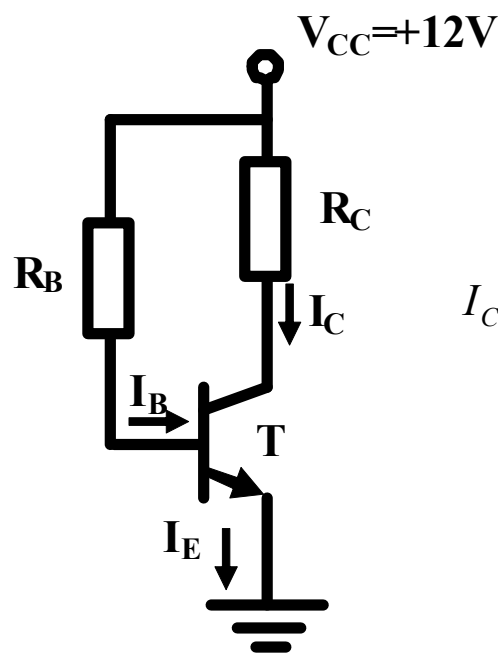
串串负反馈 灵敏度分析

$\beta=300$



	一般偏置	串串负反馈稳定输出电流	串串负反馈
输出电流 I_C	$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B} = 1.03mA$	$I_C = \frac{\eta V_{CC} - V_{BE}}{R_E + (R_{BB} + R_E)/\beta} = 1.03mA$	
$S_{V_{CC}}^{I_C} = \frac{\partial I_C}{\partial V_{CC}} \frac{V_{CC}}{I_C}$	$= \frac{\beta}{R_B} \frac{V_{CC}}{I_C} = 1.06$	$= \frac{\eta}{R_E + (R_{BB} + R_E)/\beta} \frac{V_{CC}}{I_C} = 1.30$	
$S_{\beta}^{I_C} = \frac{\partial I_C}{\partial \beta} \frac{\beta}{I_C}$	$= \frac{V_{CC} - V_{BE}}{R_B} \frac{\beta}{I_C} = 1$	$= \frac{(R_{BB} + R_E)/\beta^2}{R_E + (R_{BB} + R_E)/\beta} I_C \frac{\beta}{I_C} = 0.0167$	
$S_{R_C}^{I_C} = \frac{\partial I_C}{\partial R_C} \frac{R_C}{I_C}$	$= 0$ 未考虑厄利效应	$= 0$ 未考虑厄利效应	
$S_{R_B}^{I_C} = \frac{\partial I_C}{\partial R_B} \frac{R_B}{I_C}$	$= -\beta \frac{V_{CC} - V_{BE}}{R_B^2} \frac{R_B}{I_C} = -1$	$S_{R_{B1}}^{I_C} = \frac{\partial I_C}{\partial R_{B1}} \frac{R_{B1}}{I_C} = -0.982; S_{R_{B2}}^{I_C} = \frac{\partial I_C}{\partial R_{B2}} \frac{R_{B2}}{I_C} = 0.968$	

低灵敏度意味着高稳定性



$$S_{x_i}^y = \frac{x_i}{y} \frac{\partial y}{\partial x_i}$$

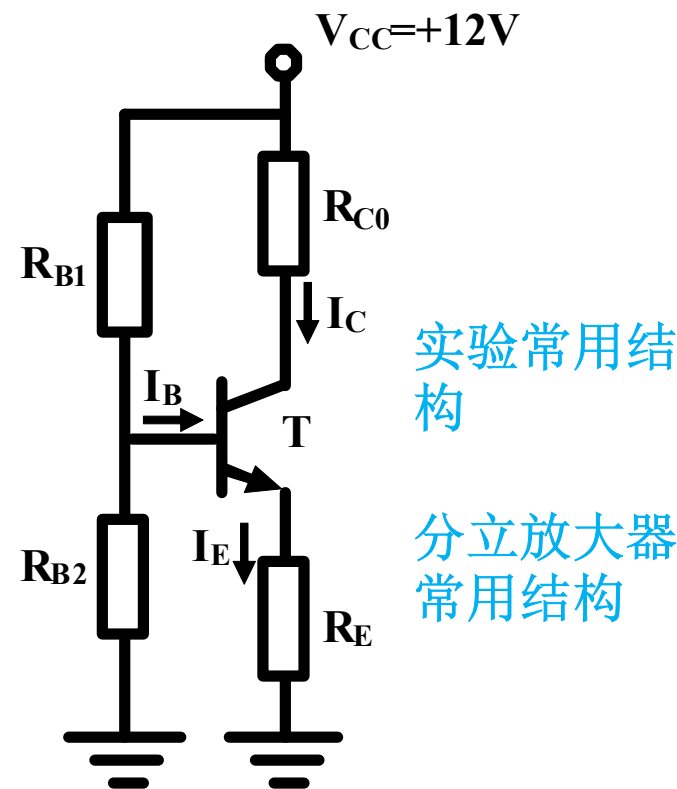
$$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B} = 1.03 \text{mA}$$

1.03mA是系统输出的设计值

$$I_C = \frac{\eta V_{CC} - V_{BE}}{R_E + (R_{BB} + R_E)/\beta} = 1.03 \text{mA}$$

$$S_{\beta}^{I_C} = \frac{\beta}{I_C} \frac{\partial I_C}{\partial \beta} = 1$$

$$S_{\beta}^{I_C} = \frac{\beta}{I_C} \frac{\partial I_C}{\partial \beta} = \frac{(R_{BB} + R_E)/\beta}{R_E + (R_{BB} + R_E)/\beta} = 0.0167$$

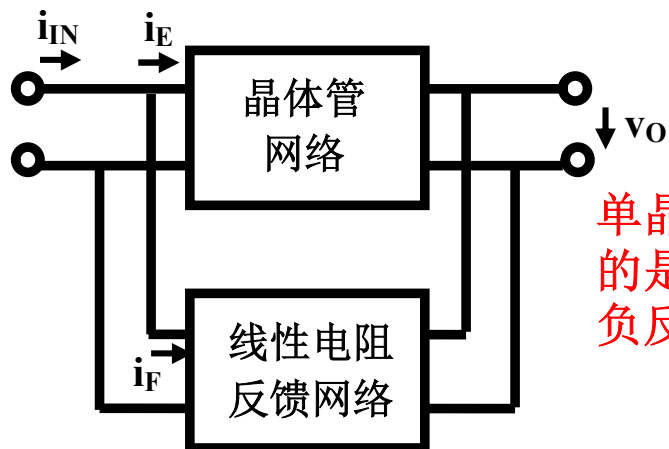


实验常用结构

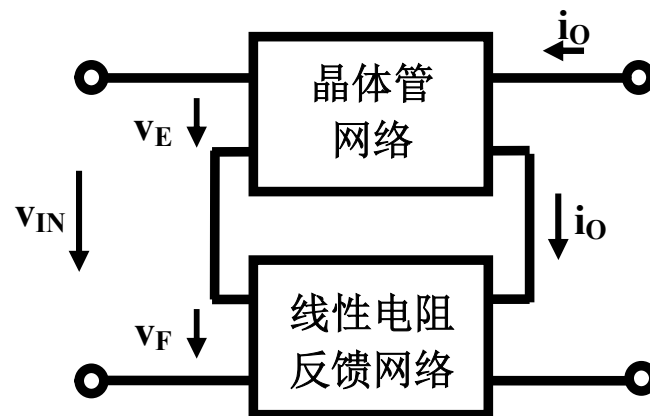
分立放大器常用结构

β : 极度不确定，设计时应确保其灵敏度很小，则系统稳定
串联负反馈使得晶体管电流增益不确定导致的直流工作点不稳定性大大降低

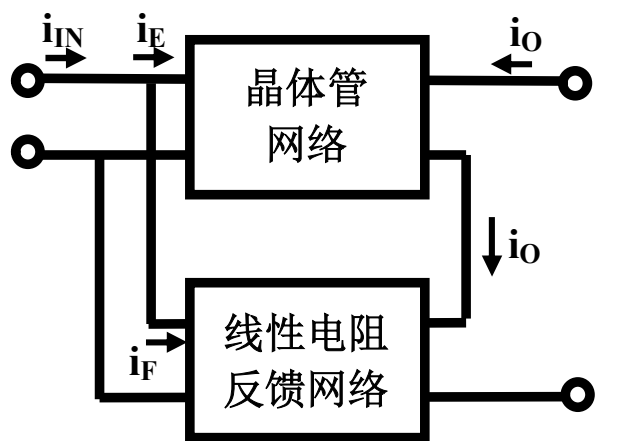
负反馈连接方式汇总



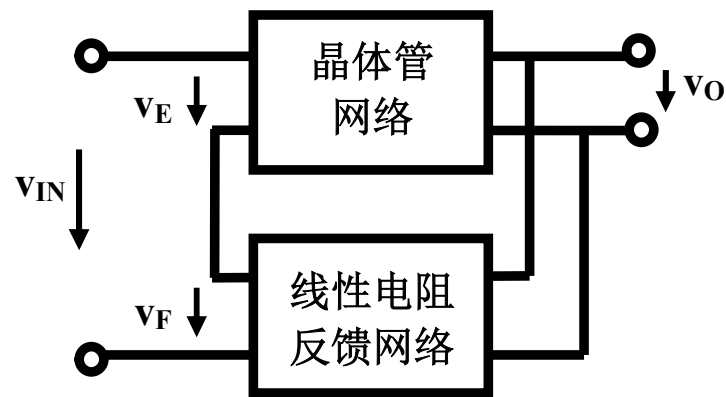
并并负反馈



串串负反馈



并串负反馈



串并负反馈

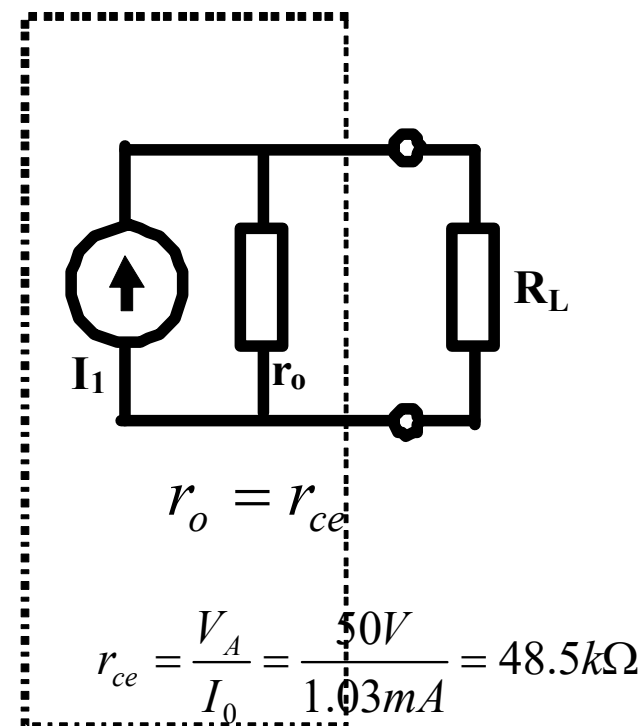
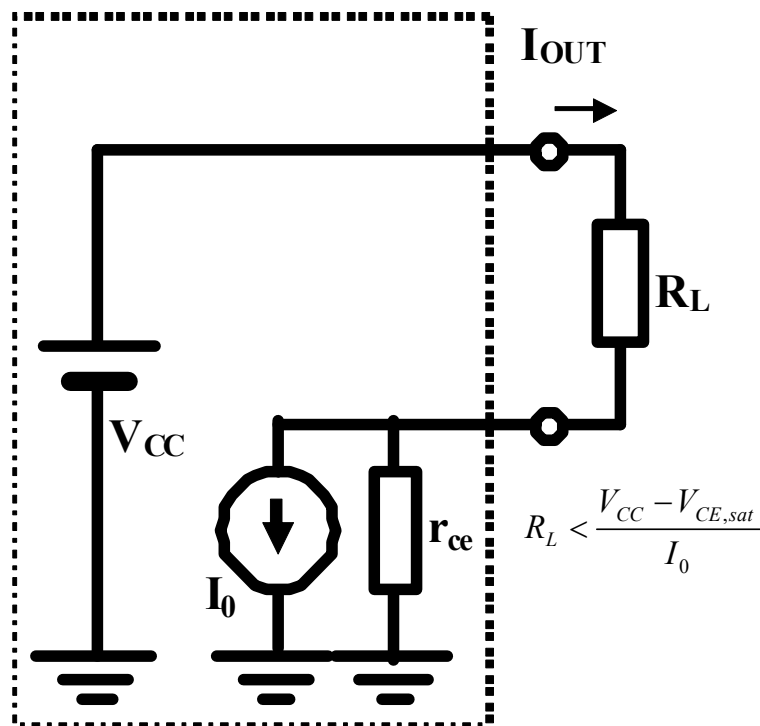
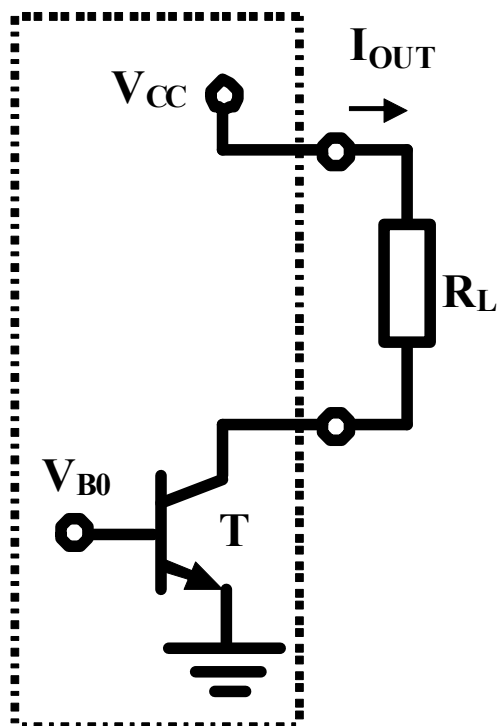
单晶体管：常见的是串串和并并负反馈

复杂晶体管网络：四种负反馈都有应用，如运放电路就是复杂晶体管网络：4种负反馈连接形成的4种接近理想的受控源

四、电流源

$$I_1 = I_0 + \frac{V_{CC}}{r_{ce}}$$

$$I_{out} = I_1 \frac{r_{ce}}{r_{ce} + R_L} = \frac{r_{ce}}{r_{ce} + R_L} I_0 + \frac{V_{CC}}{r_{ce} + R_L}$$



只要晶体管工作在恒流区，其**CE**端口对外即可等效为电流源

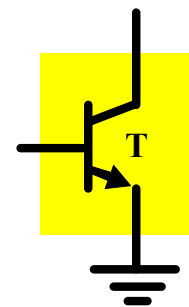
希望输出电阻越大越好
越接近理想恒流源

$$I_C = \beta A_J J_{BS0} \cdot \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \left(1 + \frac{V_{CE}}{V_A} \right) = I_0 \left(1 + \frac{V_{CE}}{V_A} \right) = I_0 + \frac{I_0}{V_A} V_{CE} = I_0 + \frac{V_{CE}}{r_{ce}}$$

负反馈改善电流源内阻特性

$$I_C = f(V_{BE}, V_{CE}) = I_{CS0} \cdot \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \left(1 + \frac{V_{CE}}{V_A} \right)$$

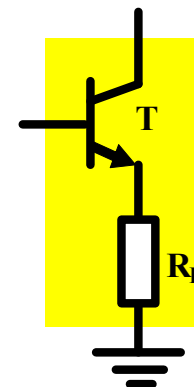
未加负反馈



$$g_o = \frac{\partial I_C}{\partial V_{CE}} = \frac{\partial f(V_{BE}, V_{CE})}{\partial V_{CE}} = \frac{\partial f}{\partial V_{CE}} = \frac{I_0}{V_A} = \frac{1}{r_{ce}} = g_{ce}$$

$$I_C = f(V_{BE}, V_{CE}) = f(V_B - I_C R_E, V_C - I_C R_E)$$

加串联负反馈电阻



$$g_o = \frac{\partial I_C}{\partial V_C} = \frac{\partial f(V_{BE}, V_{CE})}{\partial V_C} = \frac{\partial f}{\partial V_{BE}} \frac{\partial V_{BE}}{\partial V_C} + \frac{\partial f}{\partial V_{CE}} \frac{\partial V_{CE}}{\partial V_C}$$

$$= g_m \cdot \left(-\frac{\partial I_C}{\partial V_C} R_E \right) + g_{ce} \cdot \left(1 - \frac{\partial I_C}{\partial V_C} R_E \right) = -g_m R_E g_o + g_{ce} (1 - g_o R_E)$$

微分跨导增益

$$g_o = \frac{g_{ce}}{1 + (g_m + g_{ce}) R_E} \approx \frac{g_{ce}}{1 + g_m R_E}$$

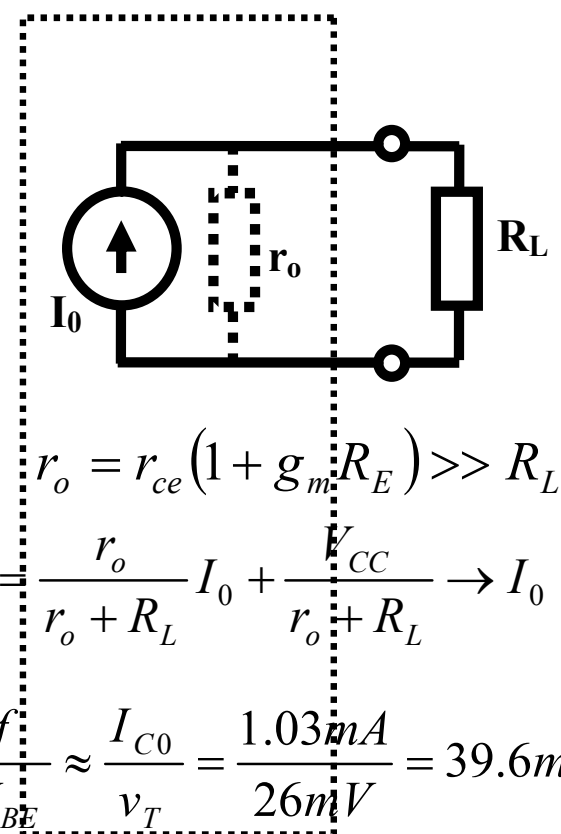
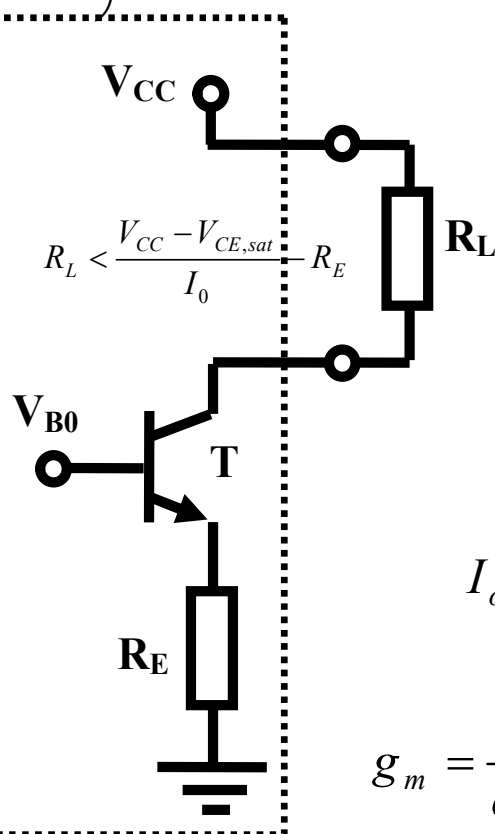
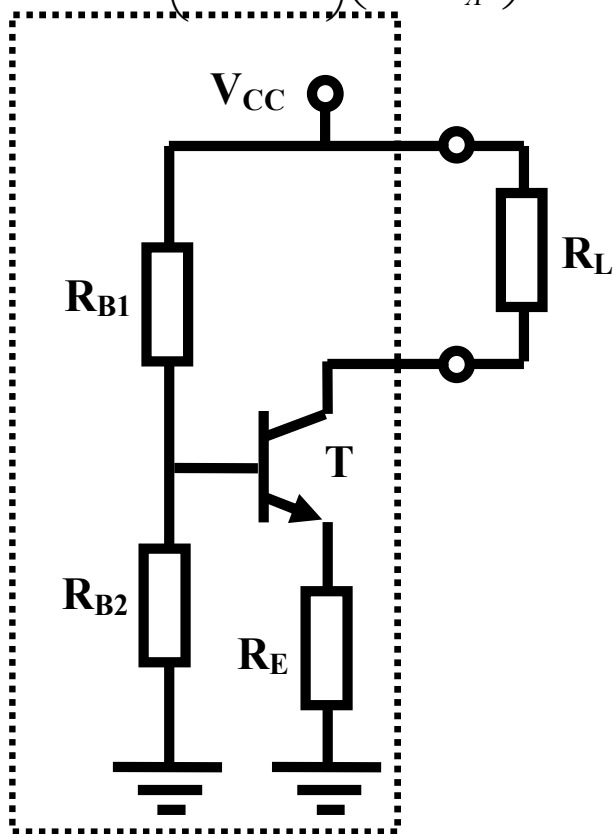
$$r_o \approx r_{ce} (1 + g_m R_E)$$

等效电流源内阻增加，
更加接近于理想恒流源

负反馈使得电流源更接近理想

$$i_C = \beta A_J J_{BS0} \left(e^{\frac{v_{BE}}{v_T}} - 1 \right) \left(1 + \frac{v_{CE}}{V_A} \right) \approx I_{CS0} \left(e^{\frac{v_{BE}}{v_T}} - 1 \right)$$

少了**2.3V**的输出电压空间



$$r_o = (1 + g_m R_E) r_{ce} = (1 + 39.6mS \times 2.2k\Omega) \times 48.5k\Omega = 88 \times 48.5k\Omega = 4.28M\Omega$$

负反馈使得温度稳定性提高

$$I_C = \beta A_J J_{BS0} \cdot \left(e^{\frac{V_{BE}}{v_T}} - 1 \right) \left(1 + \frac{V_{CE}}{V_A} \right) = I_0 \left(1 + \frac{V_{CE}}{V_A} \right) = I_0 + \frac{I_0}{V_A} V_{CE} = I_0 + \frac{V_{CE}}{r_{ce}}$$

$$I_0 = \beta A_J J_{BS0} \cdot \left(e^{\frac{V_{BE}}{v_T}} - 1 \right) \quad v_T = \frac{kT}{q}$$

$$S_T^{I_C} = \frac{\partial I_C}{\partial T} \frac{T}{I_C} \approx -\frac{V_{BE}}{v_T} \approx -27$$

$$V_{BE} = V_{B0}$$

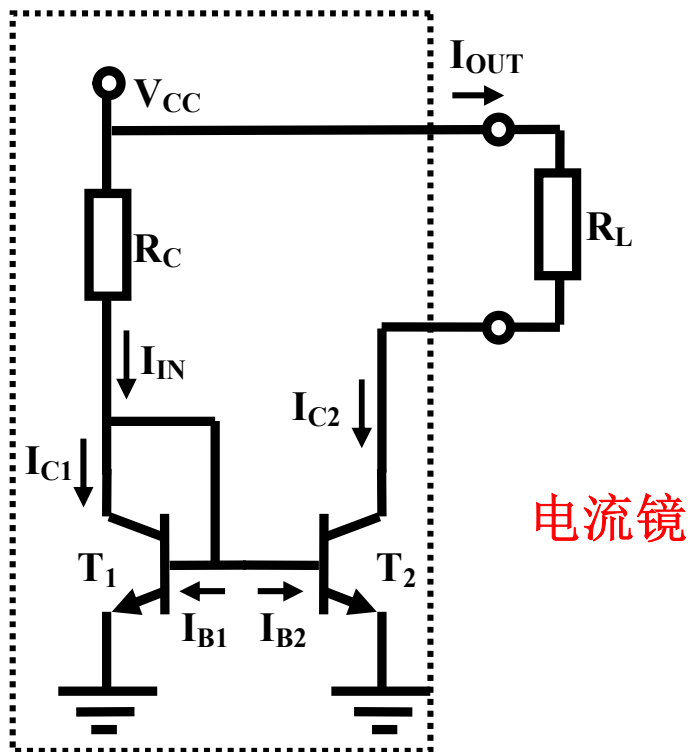
直接偏置温度敏感度极高

$$S_T^{I_C} = \frac{\partial I_C}{\partial T} \frac{T}{I_C} \approx -\frac{V_{BE}/v_T}{1 + g_m R_E} \approx -0.3$$

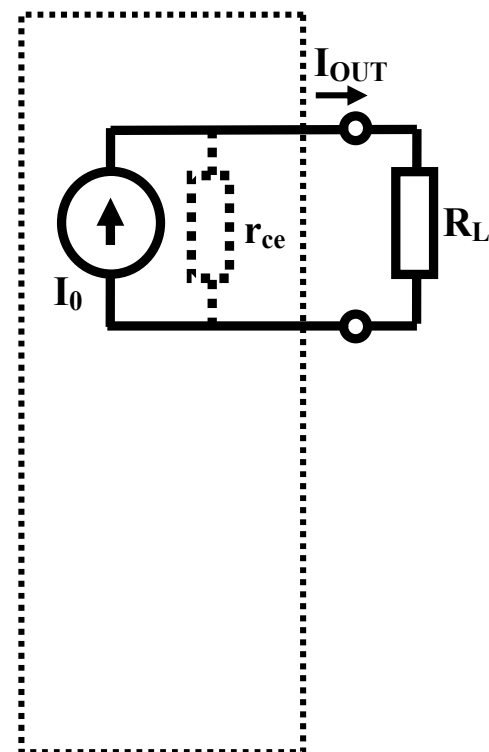
$$V_{BE} = V_B - I_C R_E$$

串串负反馈降低了温度敏感度

电流镜结构：同属性抵偿



电流镜

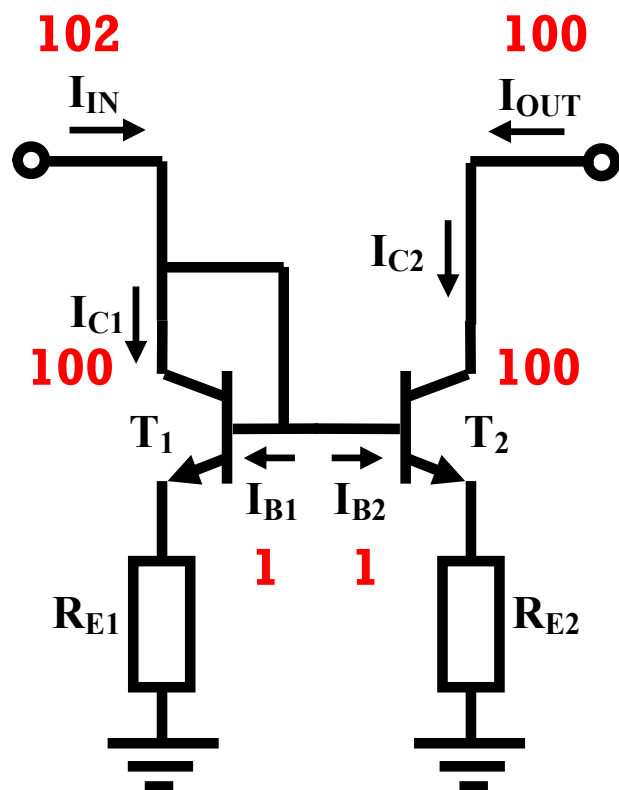


结面积
决定电
流关系

$$G_I = \frac{I_{OUT}}{I_{IN}} = \frac{I_{C2}}{I_{C1} + I_{B1} + I_{B2}} \approx \frac{I_{C2}}{I_{C1}} = \frac{\beta A_{J2} J_{BS0} \cdot \left(e^{\frac{V_{BE2}}{v_T}} - 1 \right) \left(1 + \frac{V_{CE2}}{V_A} \right)}{\beta A_{J1} J_{BS0} \cdot \left(e^{\frac{V_{BE1}}{v_T}} - 1 \right) \left(1 + \frac{V_{CE1}}{V_A} \right)} \approx \frac{A_{J2}}{A_{J1}}$$

β 越大, V_A 越大, 匹配程度越高

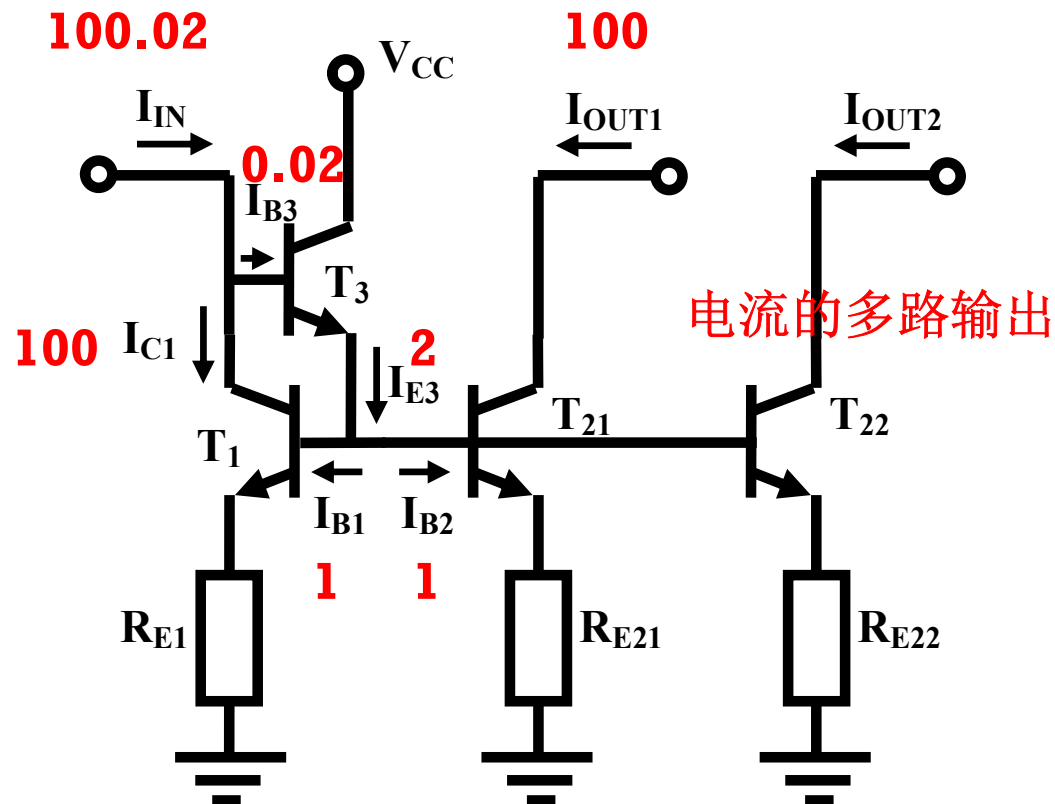
提高两条支路的匹配度



$$I_{OUT} = I_{C2}$$

$$r_o = r_{ce2} (1 + g_{m2} R_{E2})$$

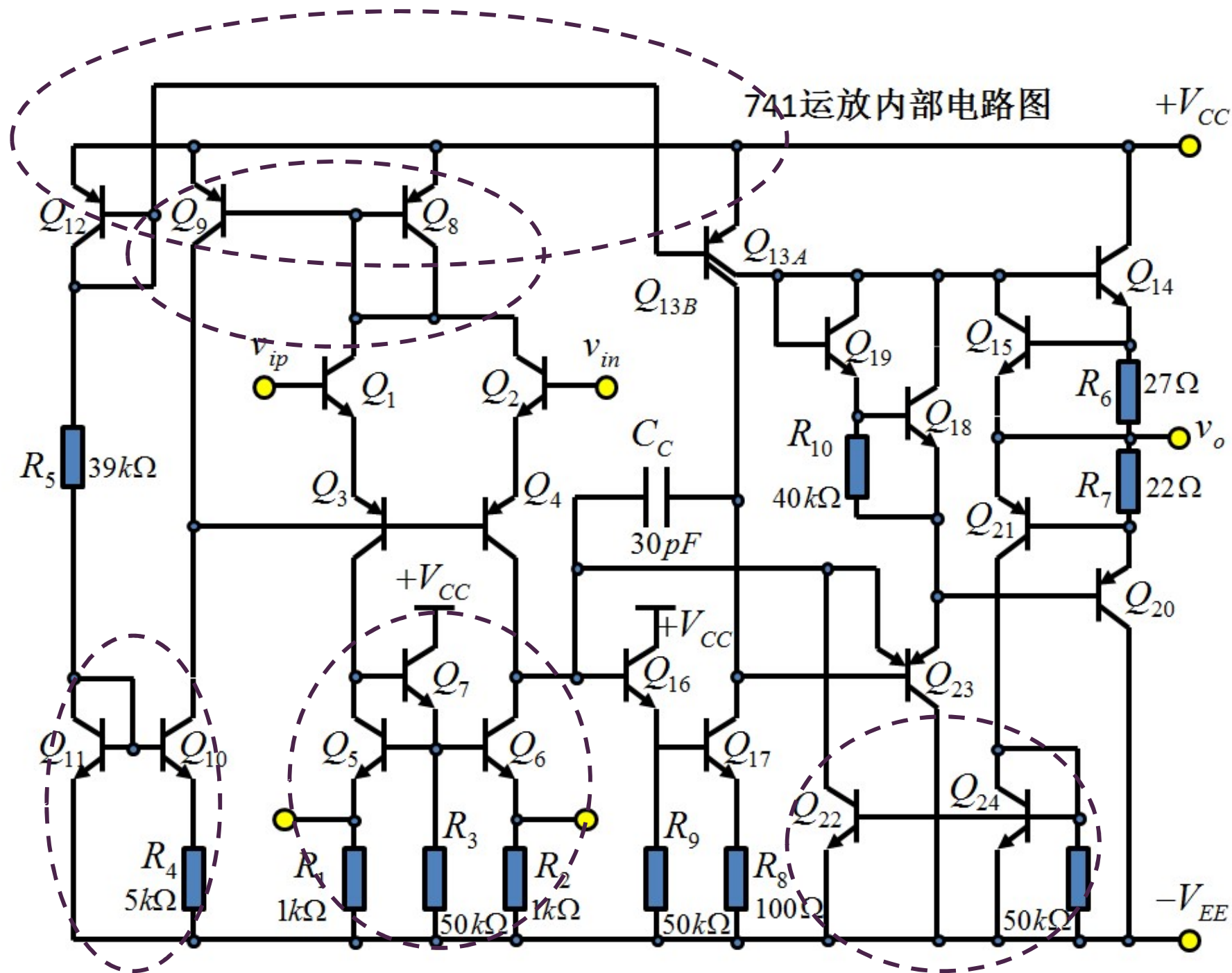
提高了输出电阻



$$\begin{aligned} I_{IN} &= I_{C1} + I_{B3} = I_{C1} + \frac{I_{E3}}{\beta_3 + 1} = I_{C1} + \frac{I_{B1} + I_{B2}}{\beta_3 + 1} \\ &= I_{C1} + \frac{1}{\beta_1(\beta_3 + 1)} I_{C1} + \frac{1}{\beta_2(\beta_3 + 1)} I_{C2} \end{aligned}$$

提高了等效电流增益

电流镜是模拟集成电路的特征电路



BJT 小结

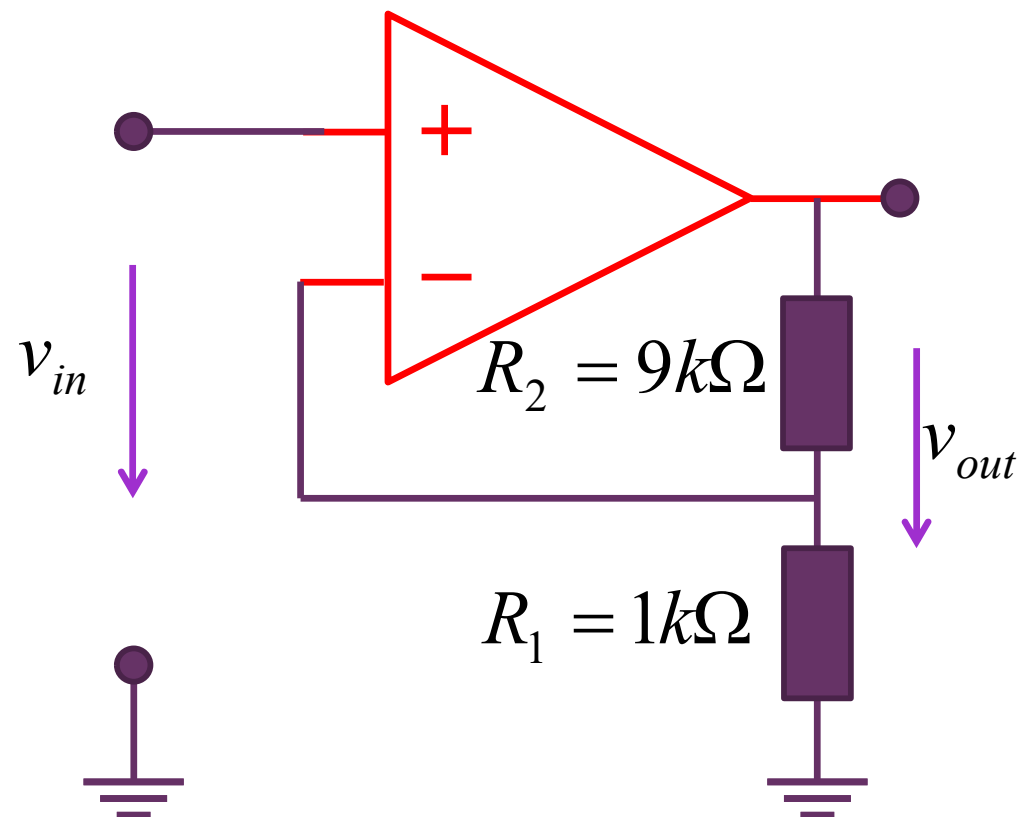
- BJT晶体管中由于同时存在两种类型的载流子电流，故称双极型晶体管
- BJT恒流区工作时，CB结反偏
 - 正常的反偏PN结截止（少子漂移电流极小，可抽象为0），但BJT中的反偏CB结可以导电，原因在于基区中有大量的发射极发射过来的来不及复合和扩散的少子（对基区而言属少子）可供漂移
 - BJT结构中的关键点在于发射极重掺杂，基区轻掺杂且很薄：发射极发射过来的少子来不及在基区复合和扩散就被（反偏CB结）集电极电压产生的电场给拉走了
- BJT做放大器使用时工作于恒流区，恒流区压控流源/流控流源模型需要把握
- 只要晶体管工作在恒流区，其受控电阻伏安特性则可等效为恒流源
 - 电流镜：用二极管做偏置，可确保输出电流和参考电流比值精确
 - 集成电路中常见的电流镜结构可以确保输出电流和参考电流之间比值由尺寸决定
 - 可以通过负反馈改善输出阻抗
 - 可以通过 β 倍增改善BJT电流镜两条支路的匹配度，MOSFET无需
 - 分压偏置：采用负反馈可稳定输出电流
 - 分压偏置电路中的串串负反馈电阻可以稳定恒流区工作晶体管的输出电流

本节课内容在教材中的章节对应

- P1001-1008: A13 BJT晶体管受控机制
- P302-321: 4.3.3 BJT分段线性化

作业6.1 运放输出电压受限于饱和电压

- 分析并画出图示电路的输入输出转移特性曲线
- 已知运放的电源电压为 $\pm 15\text{V}$ ，如果输入信号是峰峰值为 2V 的正弦波，输出是什么波形？如果输入信号为峰峰值为 5V 的正弦波，输出是什么波形？
 - 画出波形示意图

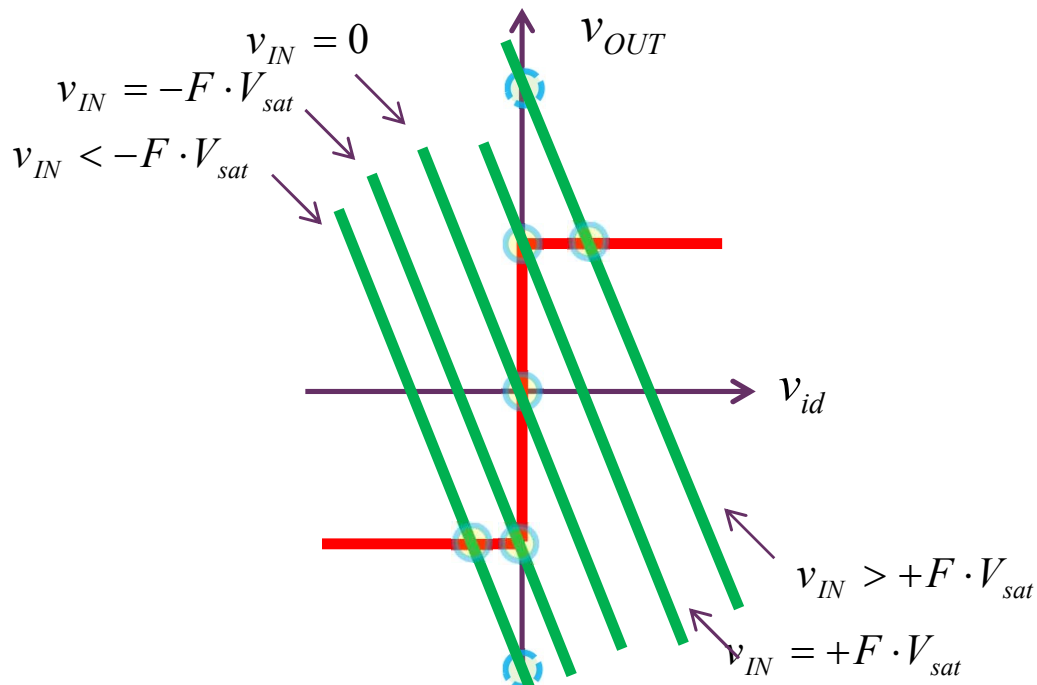
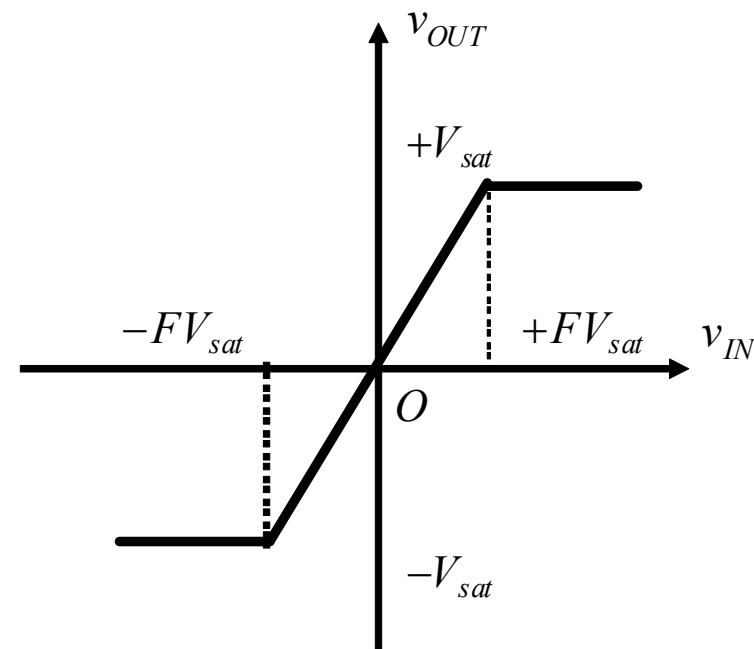
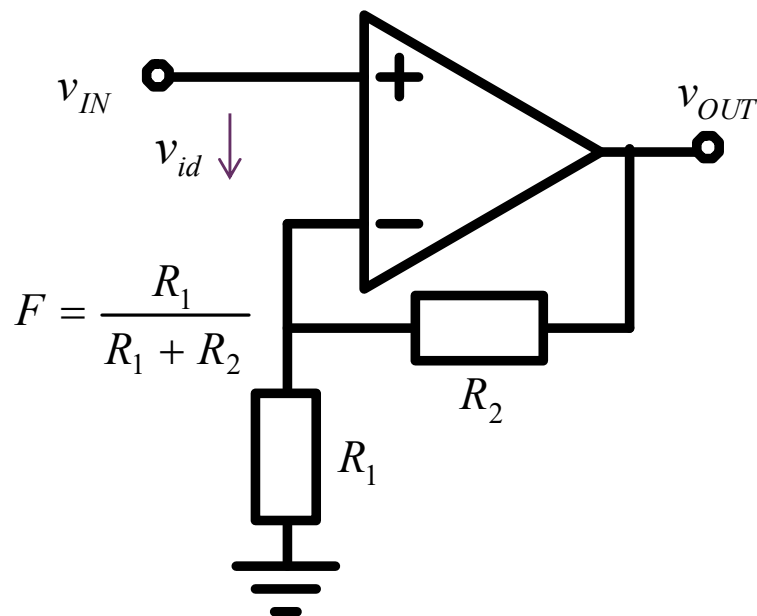


线性负反馈具有唯一解：负反馈则可假设工作在线性区

正反馈有多解：正反馈不可假设只工作在线性区

$$v_{IN} - Fv_{OUT} = v_{id}$$

$$v_{OUT} = \frac{v_{IN} - v_{id}}{F}$$

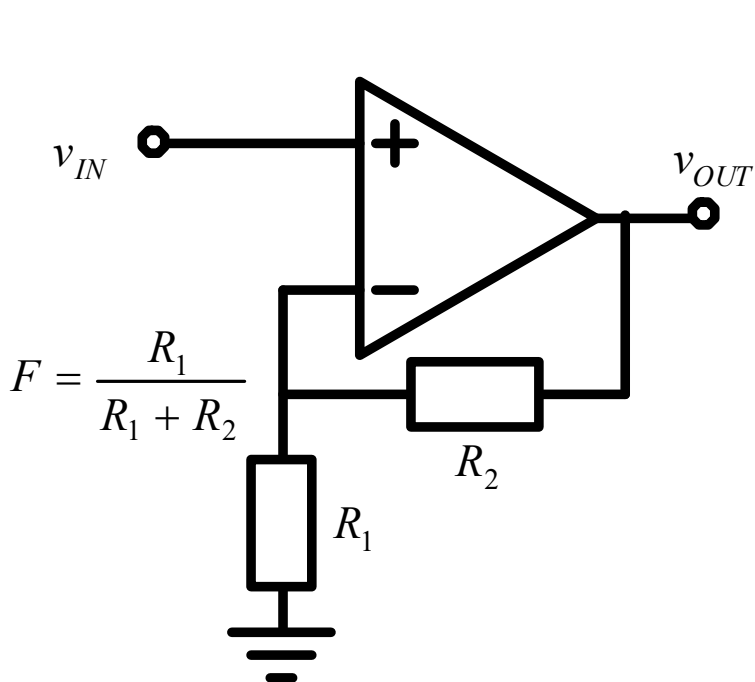


线性区 $v_{id} = 0, v_{OUT} \in [-V_{sat}, +V_{sat}]$

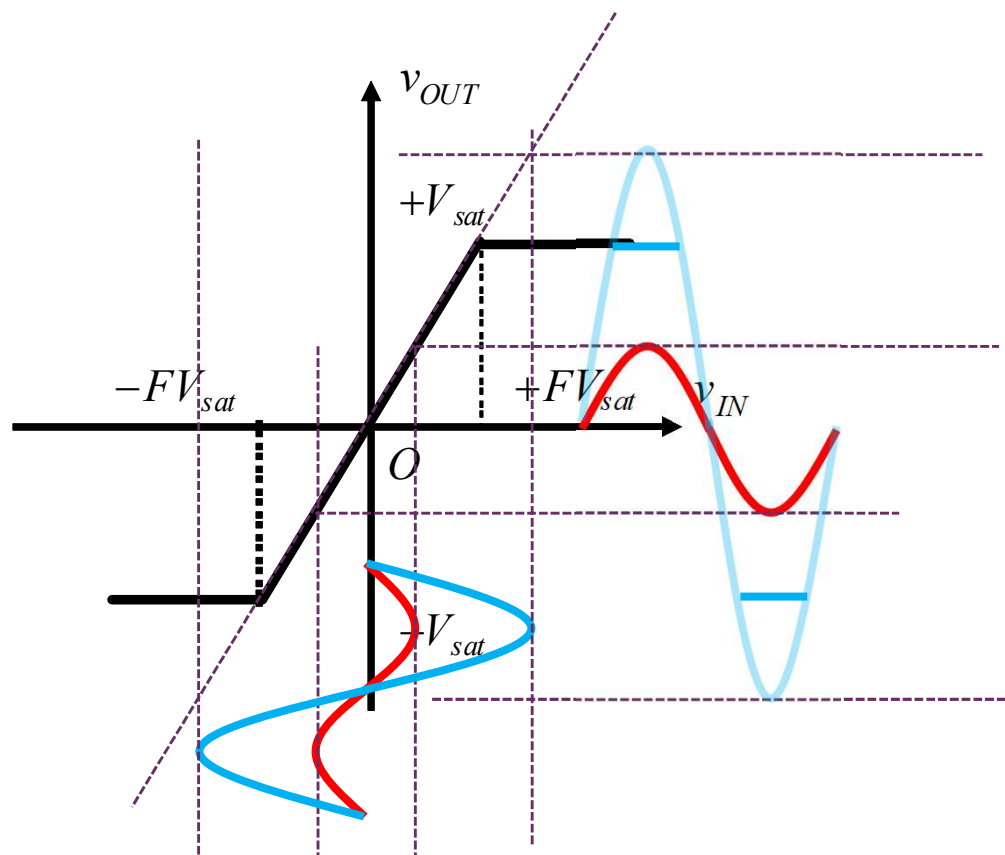
$$v_{OUT} = \frac{1}{F} v_{IN} = \left(1 + \frac{R_2}{R_1}\right) v_{IN}$$

超过饱和则饱和

- 只要确保为负反馈连接，则可假设运放工作在线性区，运用虚短、虚断原则分析，如果输出电压超出 $\pm V_{sat}$ ，则令输出为 $\pm V_{sat}$



$$v_{OUT} = \frac{1}{F} v_{IN} = \left(1 + \frac{R_2}{R_1} \right) v_{IN}$$



正解

- 已知运放的电源电压为 $\pm 15V$ ，如果输入信号是峰峰值为 $2V$ 的正弦波，输出是什么波形？如果输入信号为峰峰值为 $5V$ 的正弦波，输出是什么波形？

$$v_{IN} = V_{im} \cos \omega t$$

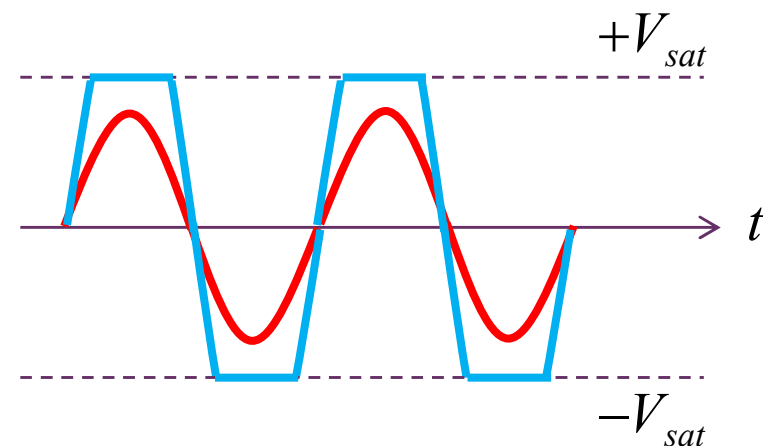
$$v_{OUT} = \frac{1}{F} v_{IN} = \left(1 + \frac{R_2}{R_1} \right) v_{IN} = 10 v_{IN} = 10 V_{im} \cos \omega t$$

$$v_{IN} = 1 \cdot \cos \omega t$$

$$v_{OUT} = 10 \cos \omega t$$

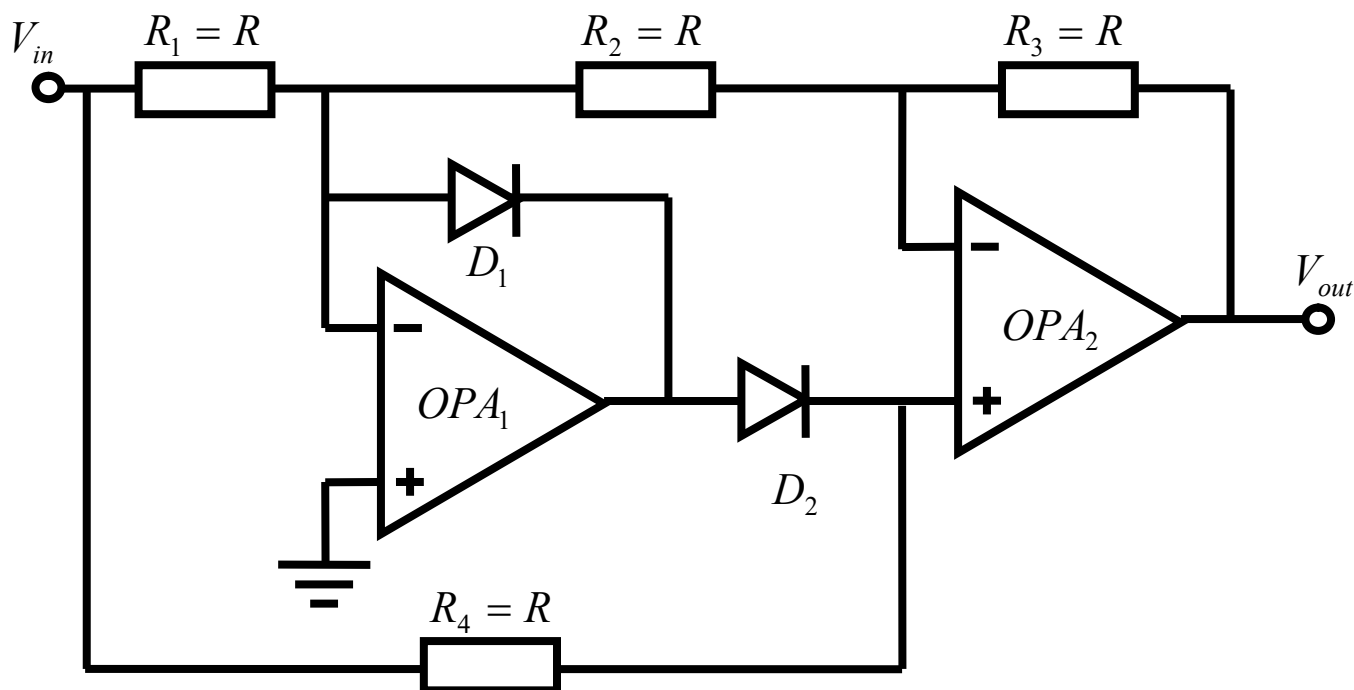
$$v_{IN} = 2.5 \cdot \cos \omega t$$

$$v_{OUT} = \begin{cases} -13V & \cos \omega t \leq -\frac{13}{25} \\ 25 \cos \omega t & -\frac{13}{25} < \cos \omega t < +\frac{13}{25} \\ +13V & \cos \omega t \geq +\frac{13}{25} \end{cases}$$



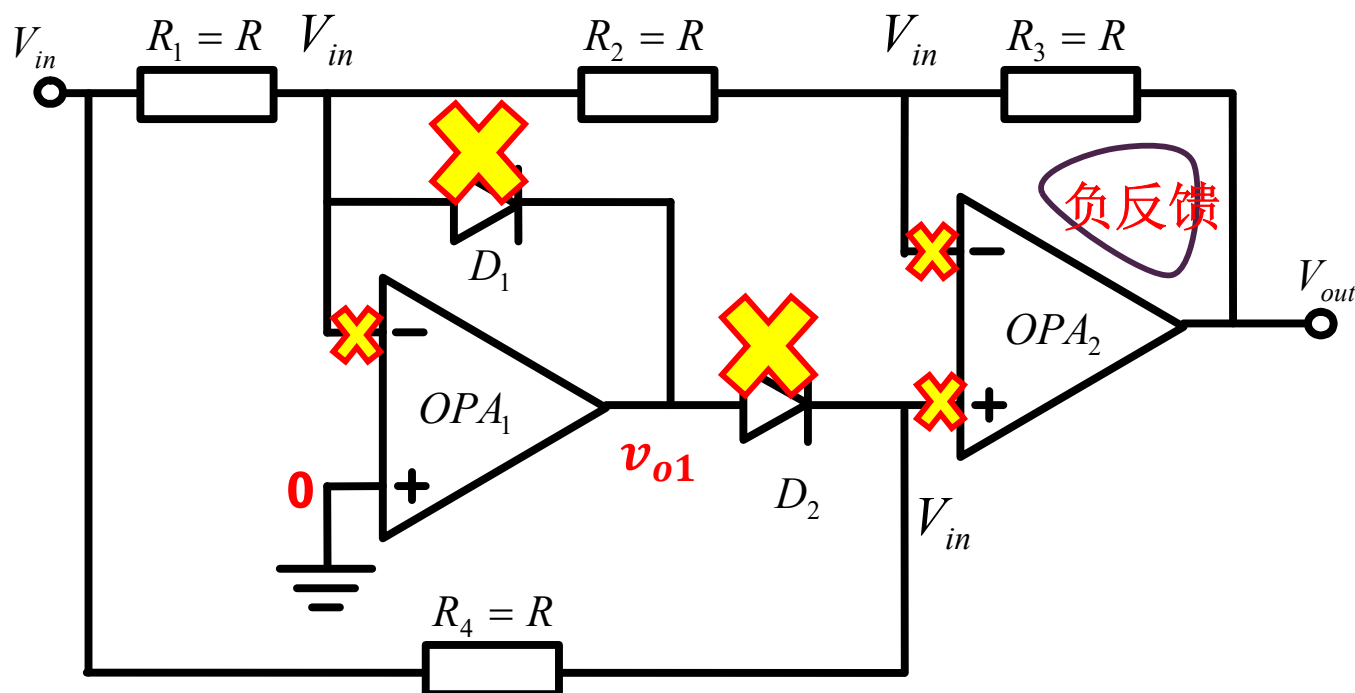
作业7.1 运放二极管电路分析

- 分析这个双运放二极管电路实现了什么电路功能？
 - 画出输入输出转移特性曲线
 - 如果输入为正弦波，输出是什么波形？



不妨假设二极管是反偏截止的

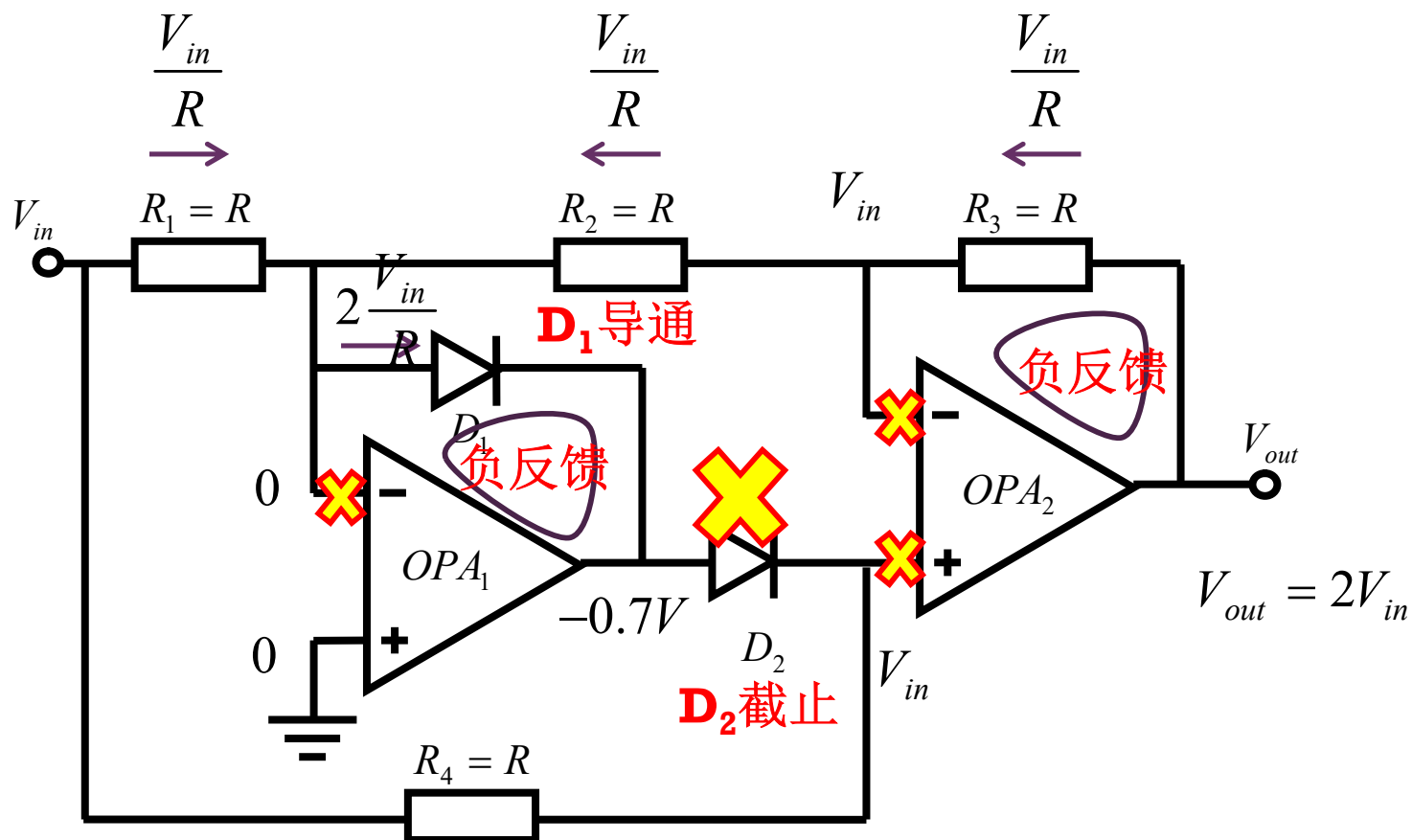
先把两个二极管抠掉，分析两端电压，再把二极管摁回去



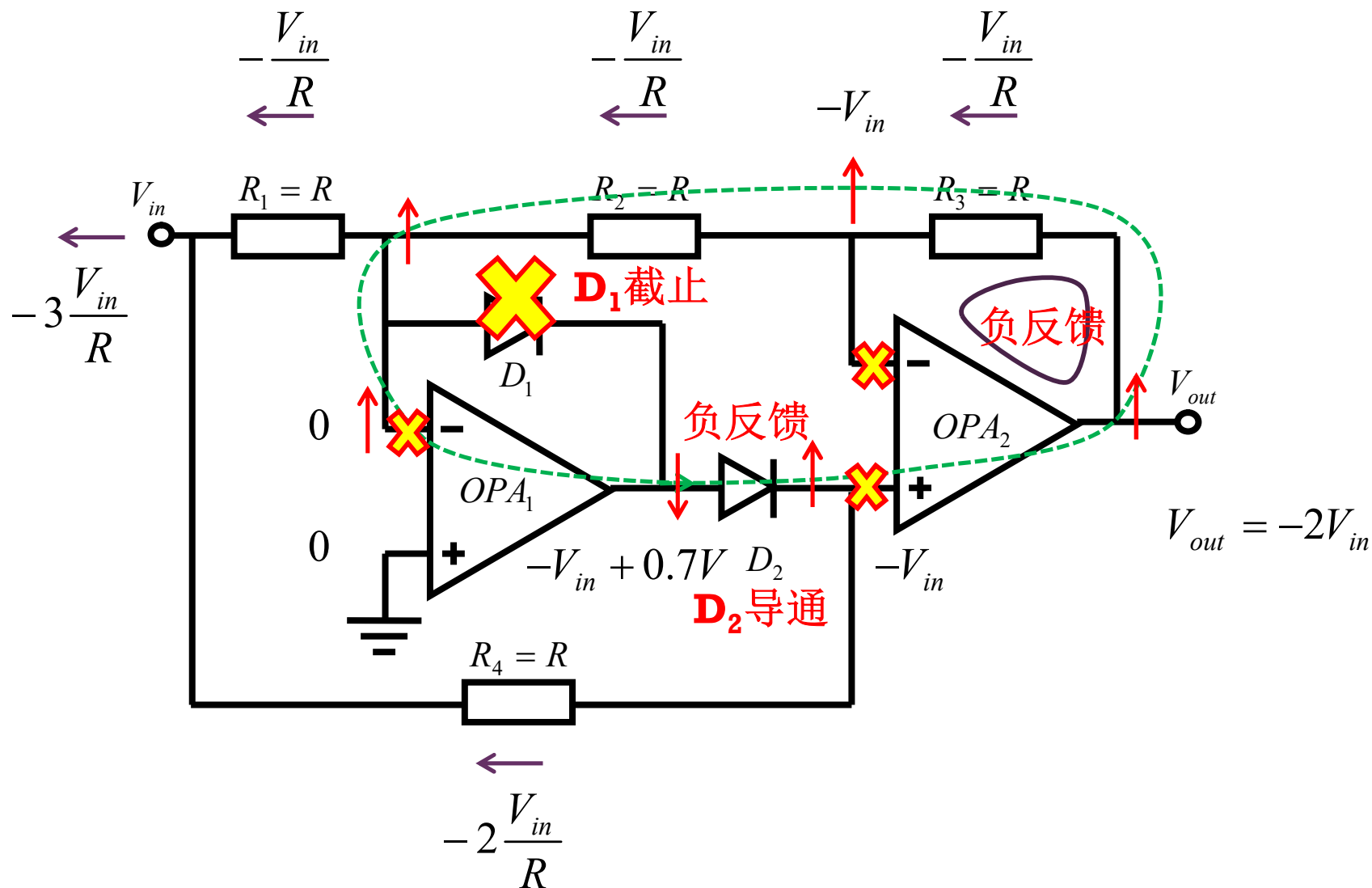
如果 $V_{in} > 0$ ， $v_{o1} = -V_{sat}$ ， D_1 反偏截止假设不成立，应假设正偏导通， D_2 反偏截止假设成立

如果 $V_{in} < 0$ ， $v_{o1} = +V_{sat}$ ， D_1 反偏截止假设成立， D_2 反偏截止假设不成立，应假设正偏导通

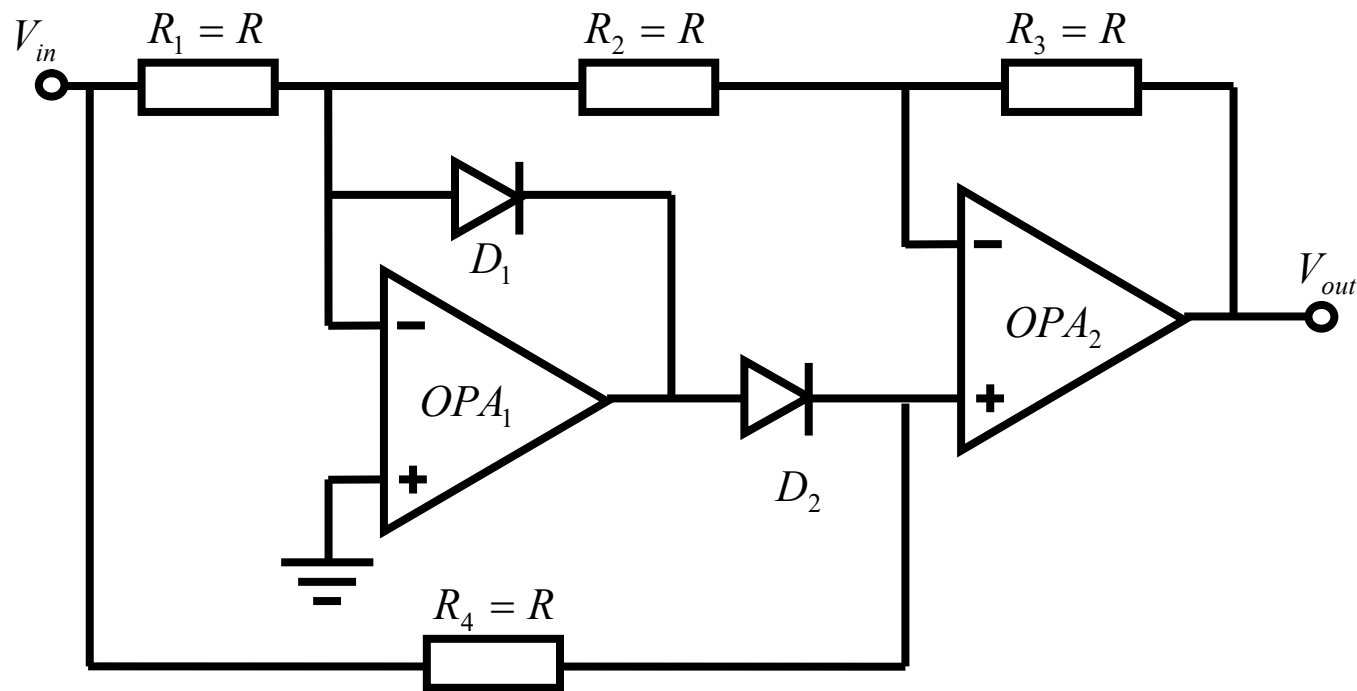
$V_{in} > 0$, D_1 导通, D_2 截止



$V_{in} < 0$, D_1 截止, D_2 导通



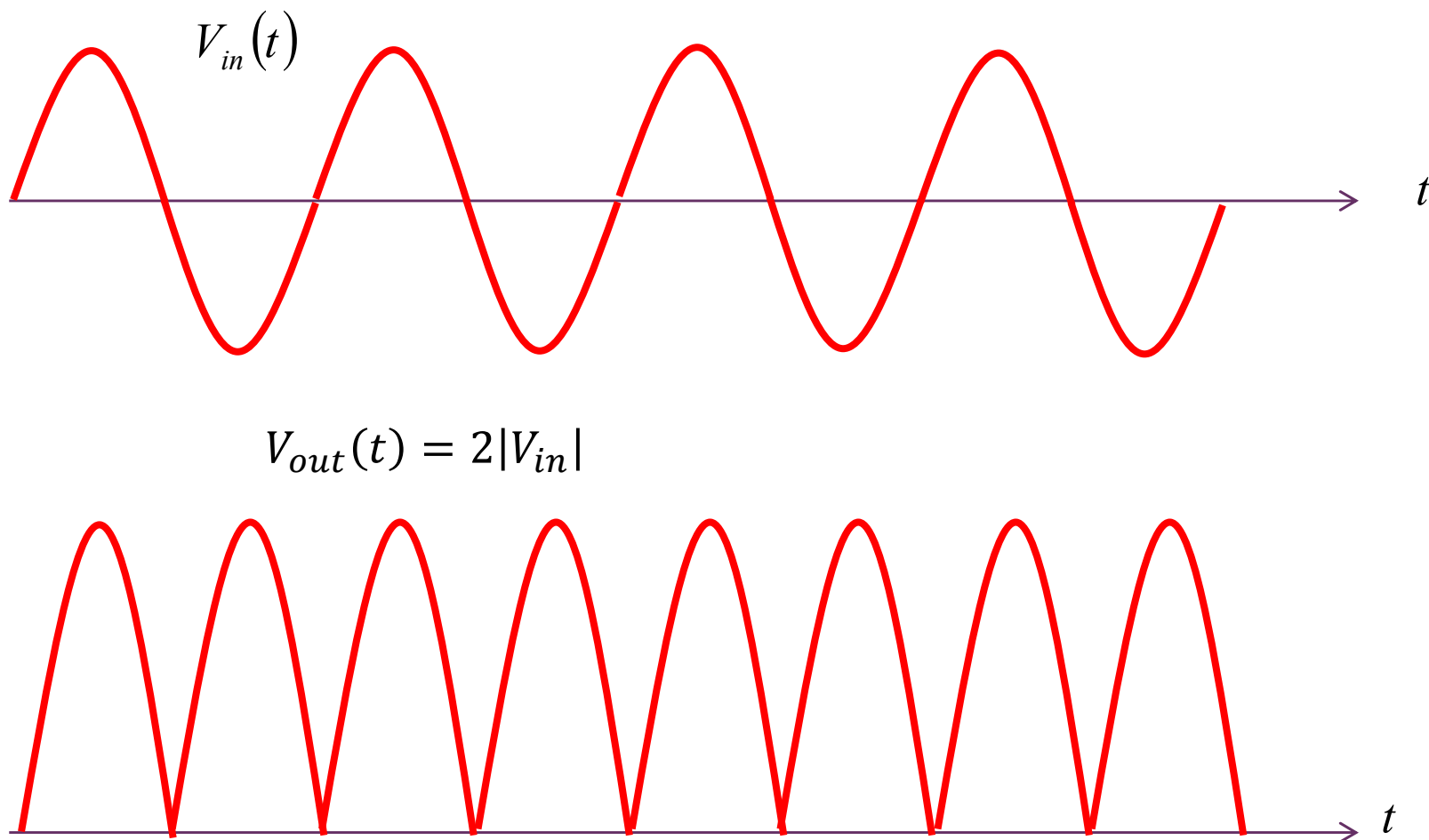
电路功能：求绝对值电路



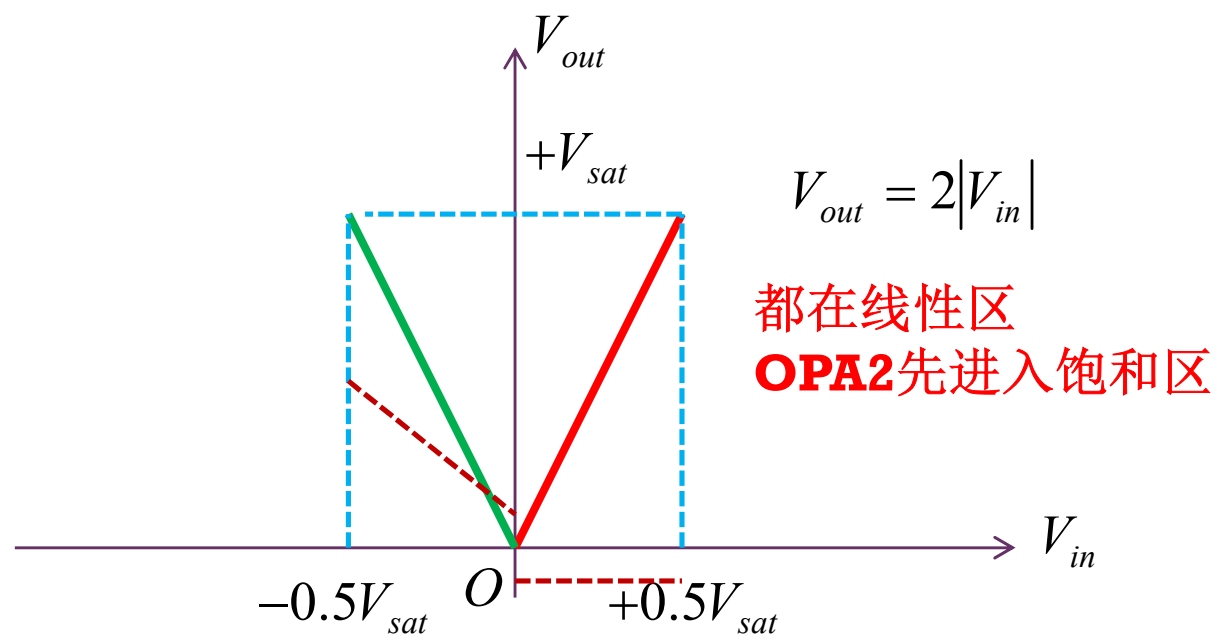
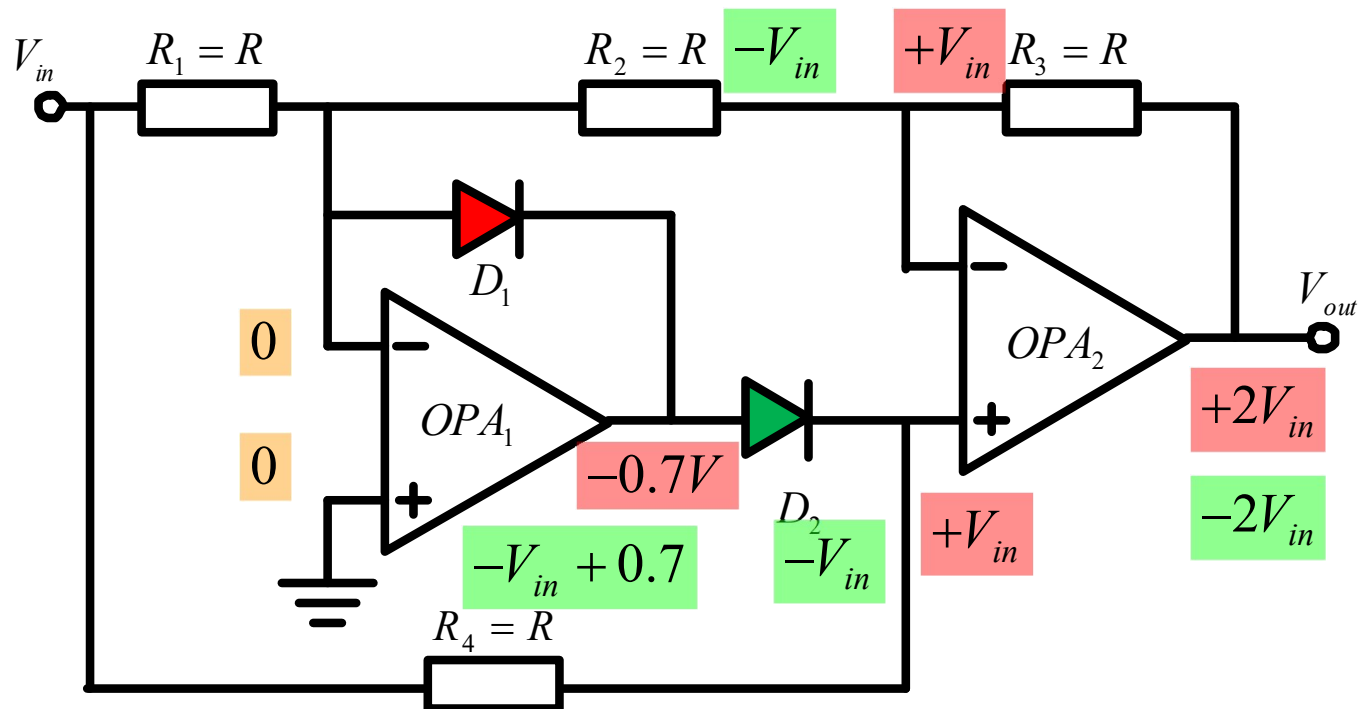
$$V_{out} = \begin{cases} +2V_{in} & V_{in} > 0 \\ -2V_{in} & V_{in} < 0 \end{cases}$$

$$= 2|V_{in}|$$

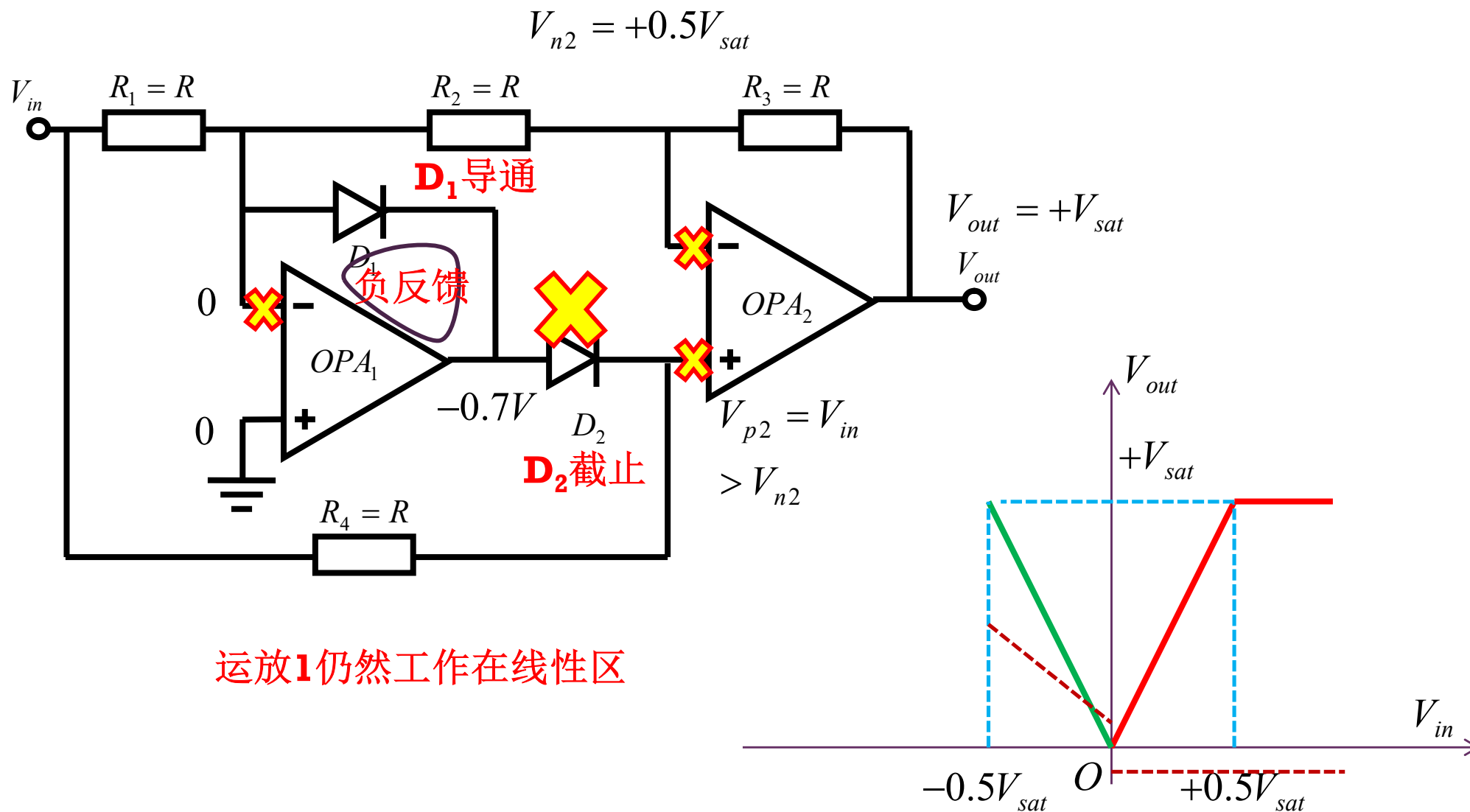
正弦激励获得全波输出



转移特性曲线



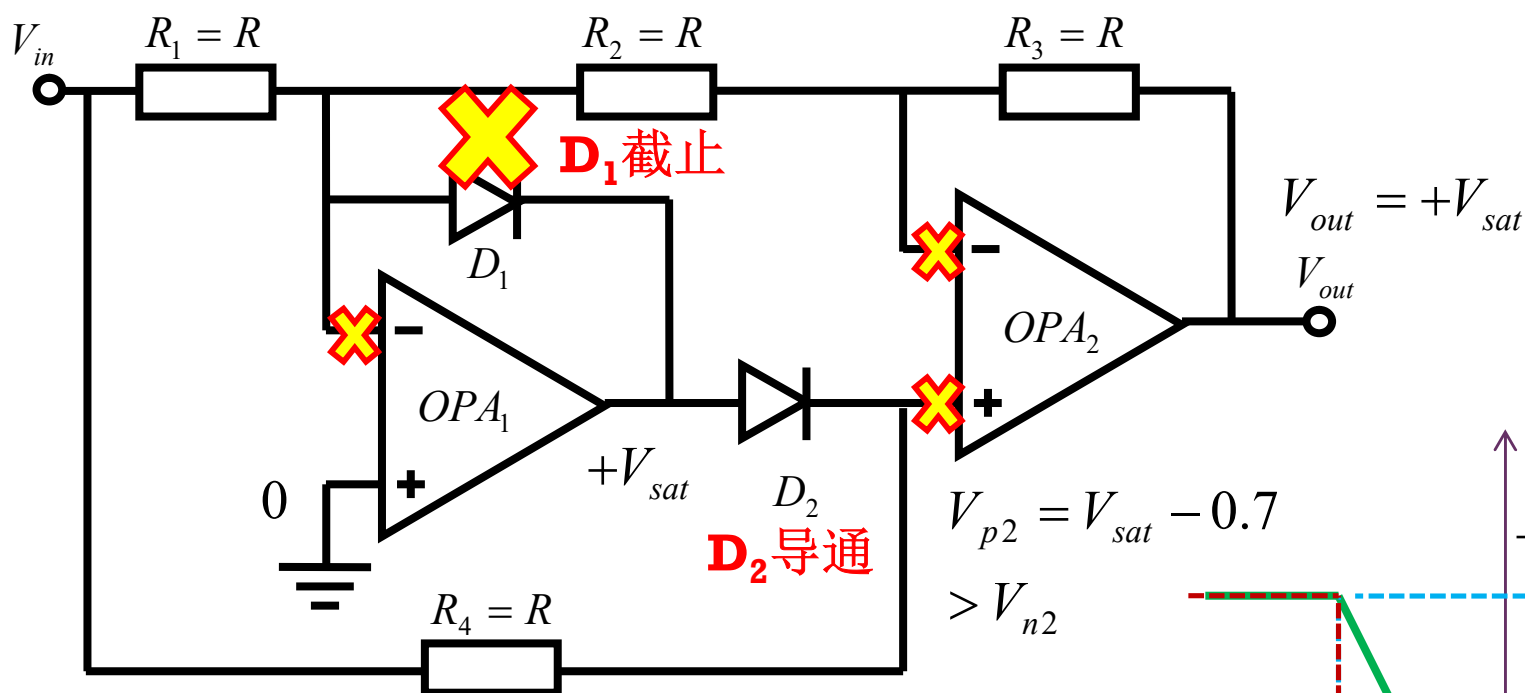
$V_{in} > 0.5V_{sat}$, OPA_2 正饱和



$V_{in} < -0.5V_{sat}$, $OPA_{1,2}$ 都正饱和

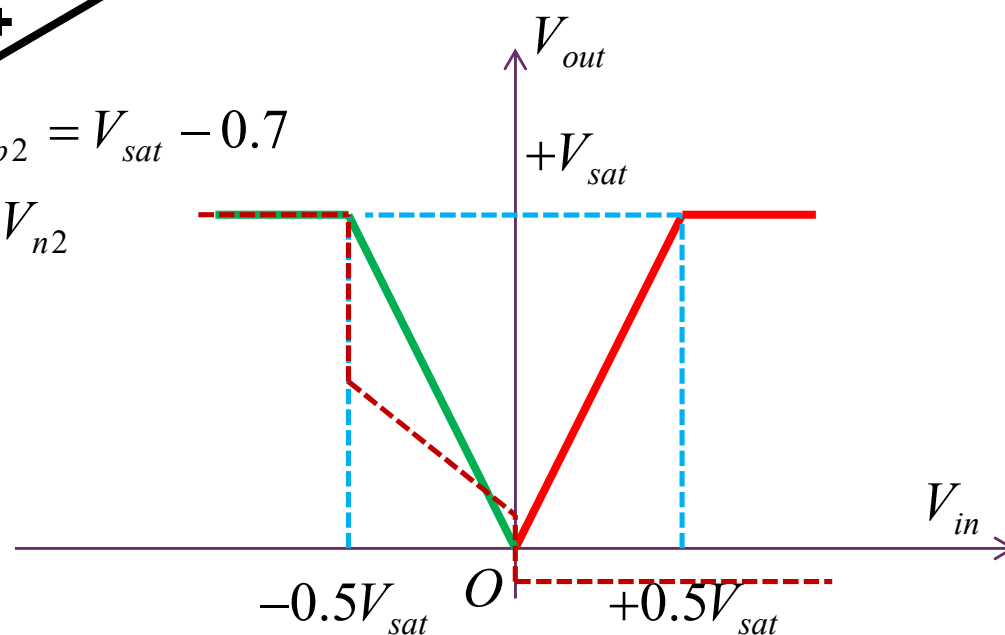
$$V_{n1} = \frac{2}{3}V_{in} + \frac{1}{3}V_{out} < 0$$

$$V_{n2} = \frac{1}{3}V_{in} + \frac{2}{3}V_{out} < -\frac{1}{6}V_{sat} + \frac{2}{3}V_{sat} = \frac{1}{2}V_{sat}$$



$$V_{p2} = V_{sat} - 0.7$$

$$> V_{n2}$$



二极管内容理解度：电路中有多个二极管无法判断二极管到底是正偏导通还是反偏截止时，可以首先假设所有二极管都是截止的（从电路中抠出来），之后分析抠除二极管的电路，判定每个二极管两端电压，如果是反偏电压则说明假设成立，仍然保持该二极管的抠除断开状态；如果正偏则说明假设不成立，则需假设该二极管是正偏导通的，用正偏导通0.7V恒压源替代。再次进行电路分析，如果确实有[填空1]流过，则说明正偏导通假设成立；如果出现[填空2]，则需再次假设该二极管反偏截止，之后再进行下一轮的分析。

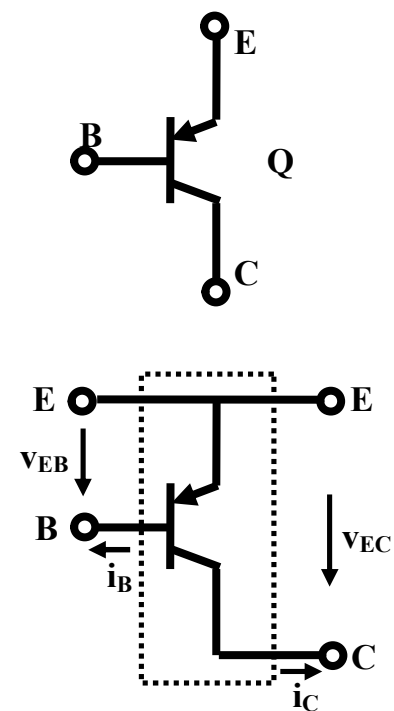
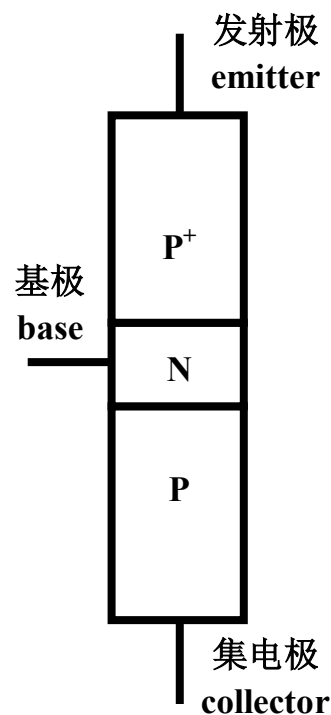
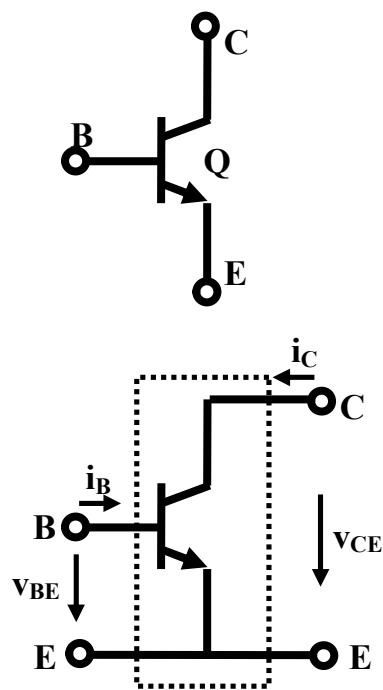
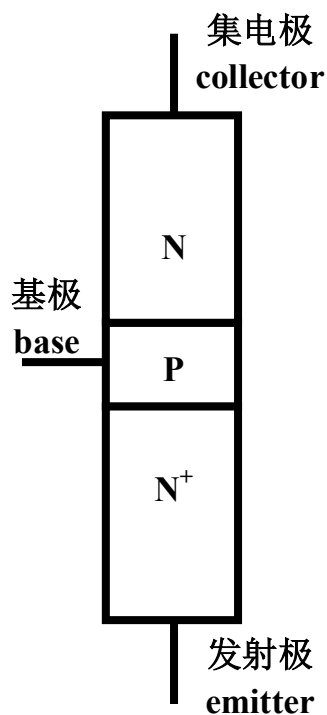
正常使用填空题需3.0以上版本雨课堂

作答

第09讲 BJT 作业

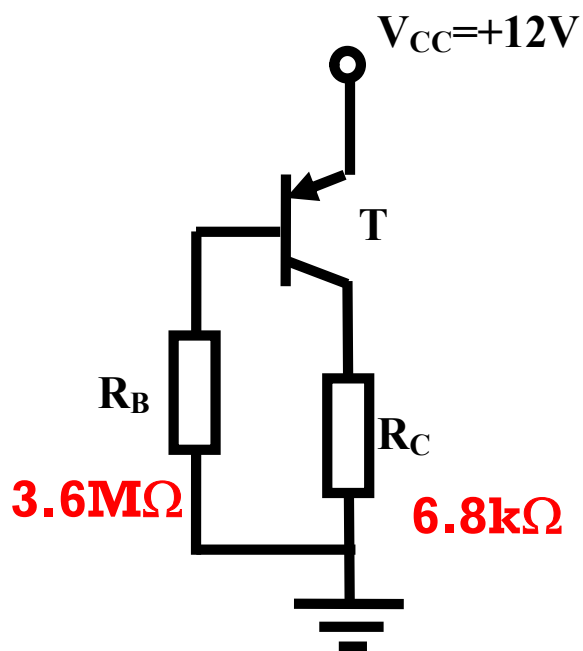
作业1 NPN和PNP

- 列表对比：（1）结构，（2）电路符号，（3）二端口定义，（4）有源区元件约束，（5）分段线性电路模型

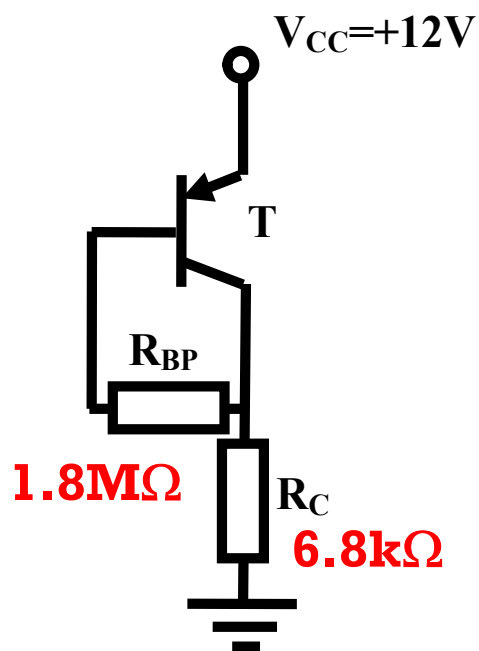


作业2 分立PNP-BJT的直流偏置

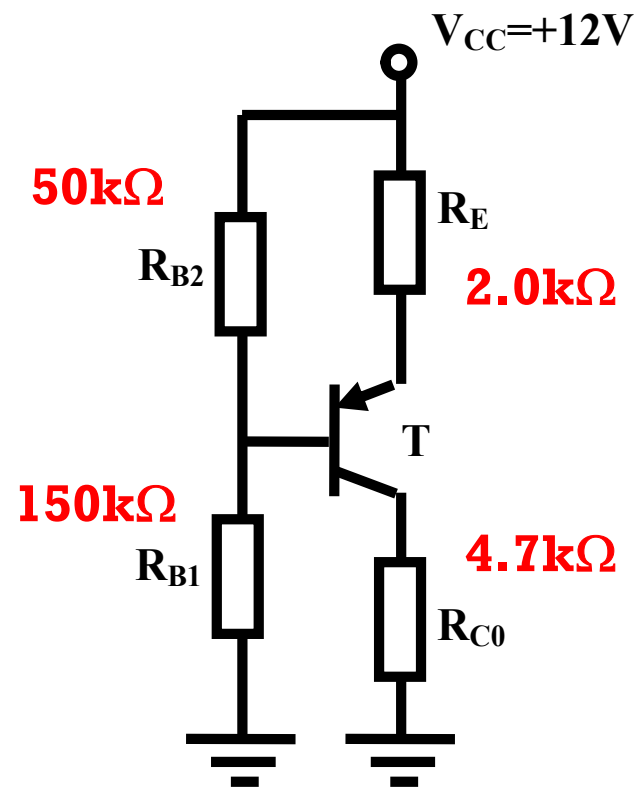
$\beta=150-400$



一般偏置



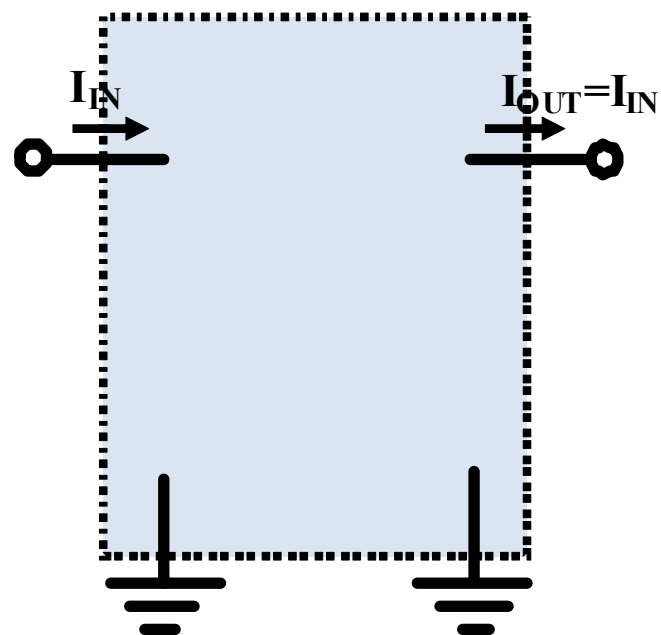
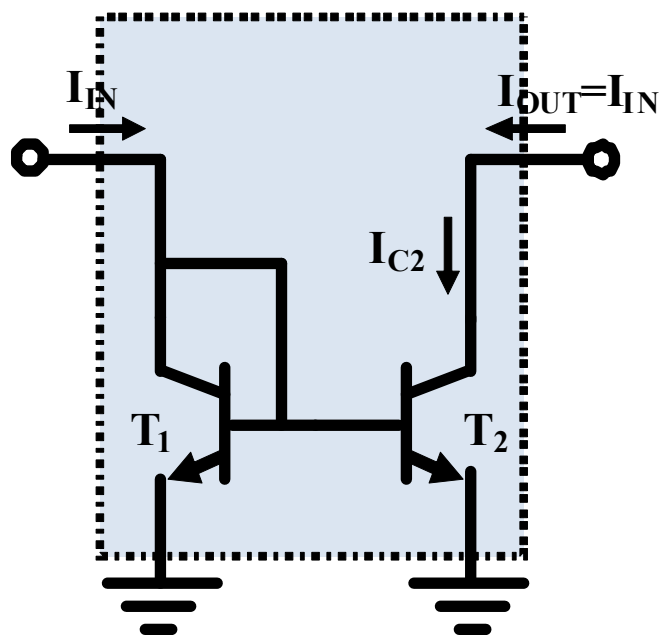
并联负反馈偏置



串联负反馈偏置
分压偏置

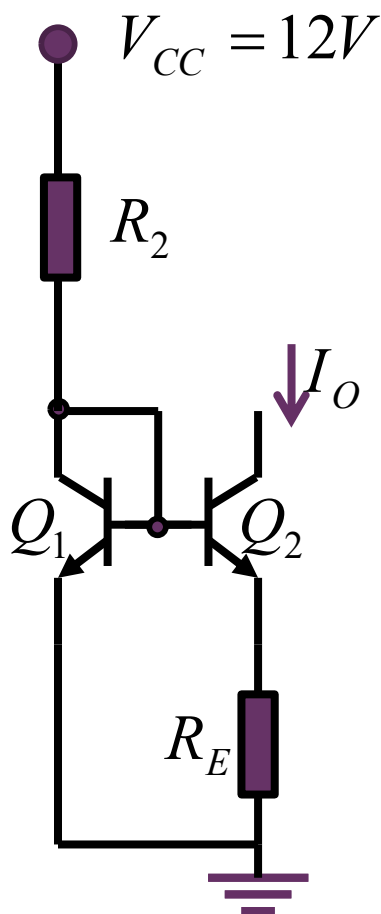
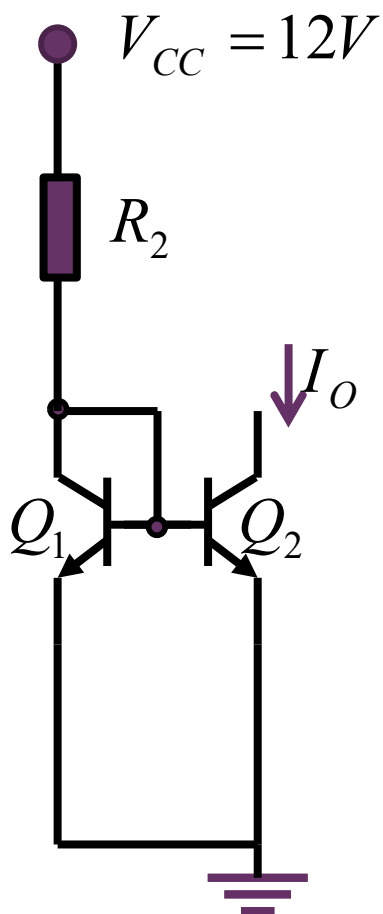
分析上述三个电路的**BJT**的直流工作点： $I_C=?V_{EC}=?$ 说明负反馈可以稳定直流工作点

作业3 电流反向



设计电路：使得流出电流等于流入电流

作业4 Widlar电流镜



- 我们希望这两个电流镜都能够产生 $12\mu\text{A}$ 的电流输出，请给出你的设计：给出电阻阻值
- 假设晶体管在 $I_C=1\text{mA}$ 时的 V_{BE} 压降为 0.7V
- （选作）说明微电流镜在产生微电流时较一般的电流源有哪些优点。

- 对于图示电路，可取 $R_B=1k\Omega$ ， $R_C=10k\Omega$ ， $V_{CC}=5V$ ，请仿真确认输入输出转移特性曲线
 - 对仿真出来的特性曲线进行分析，说明该曲线的哪个分段，晶体管位于哪个工作区？
 - 由晶体管分区电路模型分析输入输出转移特性曲线，和仿真曲线对比，偏差有多大？
 - R_B 的作用是什么？可否不要 R_B 电阻？

